# Reading the Architecture off the Weights: Manifest-Only Model Anatomy and a Census of 12,974 Hugging Face Repositories

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Engineers who deploy a model need to understand its composition, including which blocks repeat, what each block computes, how the parts are connected and where the parameters reside. This information is usually distributed across model cards, family-specific configuration files and modeling code. We show that most of it can be read from the tensor manifest alone, the list of tensor names, shapes and storage types that a safetensors file stores in a header of a few tens of kilobytes. Our method normalizes quantized storage, finds repeated blocks in the module tree, assigns roles to the parts of each block from the algebra of their shapes, and links components by matching their widths. Fields that shapes cannot determine, such as the number of attention heads, are read from `config.json` when it exists, and the configuration is checked against the weights. On 510 architectures from the `transformers` library, with ground truth taken from module classes that the method never sees, the method assigns per-tensor roles with 96.8% accuracy, compared with 82.3% for the strongest name-rule baseline, and retains 95.0% accuracy when every name is replaced by a random token. Parameter counts are exact for every model, including under NVFP4, FP8 and GPTQ storage. On 191 popular Hugging Face checkpoints read over HTTP range requests, totals agree with the Hub's own count for 149 of 158 checkpoints, with most of the differences arising because the Hub counts packed 4-bit storage, and configuration fusion recovers head geometry for 96 to 98% of models. A census of 12,974 Hub repositories, read at a median cost of 80 KB each, finds that 28% of readable Hub models are distributed without a configuration file. Among popular original language models, the share of mixture-of-experts models rose from 0% (created up to 2022) to 42% (2026), and the KV cache per token per billion parameters fell 24-fold. Quantized copies make up 6% of repositories but receive 28% of downloads, 73% of fine-tunes keep their parent's architecture, and the parameter count reported by the Hub is incorrect for 89% of NVFP4 releases. The labeled corpus and the census are released as open datasets with every raw manifest. The method runs in about 10 ms for a typical model, and we release it as an open explorer that can analyze any Hugging Face repository without downloading its weights.

**Date:** November 2026
**Playground:** https://joessattes.github.io/ai-model-explorer/
**Hub census:** https://joessattes.github.io/ai-model-explorer/census/
**Datasets:** https://huggingface.co/datasets/JoesSattes/hub-weight-census
https://huggingface.co/datasets/JoesSattes/manifest-anatomy-corpus
**Code:** https://github.com/JoesSattes/ai-model-explorer
**Python:** `pip install ai-model-explorer`



**AI Model Explorer**

## 1 Introduction

Deploying, compressing or auditing a pretrained model requires knowledge of its architecture: the number and types of its blocks, whether it uses mixture-of-experts layers and how many of its parameters they hold, and which parts are executed once per input rather than once per generated token. For established model families, this information is available in the configuration file (`config.json`); for new models, it may be incomplete or ambiguous. Configuration keys differ between families (`hidden_size`, `d_model`, `n_embd`), composite models nest several schemas, and some fields, such as `layers_block_type`, can

be interpreted only together with the modeling code. Inspecting that code is time-consuming, and executing it through `trust_remote_code` is a documented security risk on public model hubs [47]. Graph viewers such as Netron [35] and Model Explorer [15] require an exported computation graph and, in practice, the complete weight file.

However, a much smaller description of a checkpoint is available in almost every case. A safetensors file [23] begins with an 8-byte length field followed by a JSON header that records the name, data type, shape and byte offsets of every tensor, and two HTTP range requests suffice to retrieve it. Across the 510 architectures studied here, the median header is 46 KB, about $6 \times 10^{-5}$ of the size of the weights in BF16. PyTorch zip checkpoints store equivalent information in their `data.pkl` record, which can be parsed without reading tensor data. We refer to this list of tensor names, shapes and storage types as the *tensor manifest*.

This paper studies how much of a model's architecture can be recovered from the tensor manifest alone, and how reliably. The question is motivated by practice, since a method that answers it can describe any model on a public hub before its weights are downloaded. It is also of analytical interest, because some architectural properties leave no trace in tensor shapes. A recovery method should therefore identify these properties, quantify how often a default assumption about them holds, and obtain them from a more reliable source when one is available.

Our approach rests on the observation that a manifest carries two kinds of evidence. The name tree reflects structure: integer-indexed lists correspond to stacks of blocks, identical subtrees to identical block types, and the module that owns a stack to a component. The shapes, in turn, reflect function. All blocks in a stack share a residual width $d$. An attention region consists of several projections that map from $d$ and one that maps back to it; a mixture of experts is a group of tensors whose leading dimension equals the output size of a router; a selective state-space mixer is a depthwise convolution accompanied by a few per-head vectors; and two components are connected when the output width of one equals the input width of the other. Names are used only to resolve the remaining ambiguities; as Section 5.2 shows, most of the recovery is preserved even when every name is replaced by a random token.

The contributions of this paper are as follows. (1) We define manifest-only architecture recovery, propose a representation of a model's anatomy suited to engineering use, and analyze which of its properties tensor shapes cannot identify (Section 3.2). (2) We present a recovery algorithm that combines module-tree structure with shape algebra and width matching, handles eight quantized storage layouts, and fuses non-identifiable fields from the configuration while checking the configuration against the weights (Section 3). (3) We evaluate the algorithm on 510 architectures, with ground truth derived from module classes, and on 191 popular checkpoints read directly from the Hugging Face Hub. We also release it as a bilingual web tool and a Python package (Sections 4–5.4). (4) We apply it to a stratified census of 12,974 Hugging Face repositories, taking architectures from the weights rather than from declared metadata, and report how architectures changed by creation year, how derivatives relate to their parent models, how downloads are distributed, how often metadata disagrees with the weights, and what fraction of each year's releases fits common devices, with confidence intervals and trend tests throughout (Section 5.5). The census and the labeled corpus are released with every raw manifest, together with an open board to which anyone can submit a model. The appendix reports the results for every model.

## 2 Related work

A checkpoint usually comes with three descriptions of the same model: the weights, a configuration file and code. Model cards [29] add a human summary whose architectural fields are free text. The configuration is authoritative for the library that wrote it, but its schema depends on the family and, for new families, has to be interpreted through code. Pickle-based weight formats can execute code when loaded [47]. Safetensors avoids

this risk, and its header is, to our knowledge, the most uniform self-description available across families. We therefore treat this header as the primary source of information.

Reading safetensors headers is itself an established practice. The Hugging Face Hub fetches safetensors headers with range requests to show parameter counts per data type [24], and several viewers list tensor names, shapes and types. hf-mem [3] uses the same requests to estimate inference memory for the weights, adding a KV-cache estimate taken from `config.json`. HF Viewer [10] draws architecture graphs for Hub models from their configuration and PyTorch module layout. These tools report sizes without saying what each tensor does, or obtain the structure from configuration and code. Our contribution lies in interpreting the header, that is, in recovering components, block types and tensor roles from the manifest alone, so that it applies to families no tool has been written for, to checkpoints whose configuration is missing or wrong, and, as the shapes-only experiment in Section 5.2 shows, without relying on names.

The closest existing practice is the name tables that conversion tools maintain, such as llama.cpp's mapping from Hugging Face module paths to its own tensor types [14]; these tables are precise for the architectures they list but do not cover others, and we use one of them as a baseline. Netron [35], TensorBoard and Model Explorer [15] draw operator graphs from exported formats. They show a graph faithfully but do not group operators into semantic parts, and they need the full file. Transformer Explainer [4] explains one fixed architecture interactively. Our tool adopts its step-by-step presentation but derives the content from the manifest, so that it can be applied to any architecture.

Large-scale studies of the Hub map lineage, licenses and documentation from repository metadata, model cards and base-model tags [25, 39]; the Model Atlas [21] charts model relations and predicts undocumented attributes, starting from the documented part of the Hub. These studies reveal the extent of missing documentation on the Hub. Our census examines the same ecosystem from a complementary perspective, namely the information contained in the weights themselves, which is available even when documentation is absent, and the frequency with which the documentation disagrees with it.

Prior work has studied architecture recovery mainly in an adversarial setting. Side-channel methods reconstruct a network from cache timing or memory traces of a running model [22, 44]; in that setting the observer has no access to the file. In our setting the header is public, and the difficulty lies in interpreting it. Weight fingerprinting such as HuRef [46] identifies a model's lineage from weight values, which we never read.

Our shape rules encode the signatures of attention and its variants, including grouped-query [1] and latent attention [7], sparse mixtures of experts [12, 37], selective state-space and linear-attention mixers [6, 17, 45] and their hybrids [30], conformers [18, 34], transducers [16, 43], diffusion transformers [32] and tabular in-context learners [20, 33]. The storage rules cover GPTQ [13], AWQ [26], FP8 [28], MXFP4 [36], NVFP4 [31] and the MLX [19] and bitsandbytes [8] layouts.

# 3 Method

This section defines the problem, gives an overview of the system, and then describes the recovery algorithm and the configuration fusion step in turn.

## 3.1 Problem formulation

A manifest is a set $\mathscr{M} = \{(n_i, s_i, \tau_i)\}$ of tensor names (dot-separated paths), shapes and storage dtypes. A configuration $\mathscr{C}$, a nested key–value tree, may also be available. The goal is to recover an anatomy $\mathscr{A}$ with four levels: components, stacks of blocks, block types and regions. At the top are components, such as a vision tower, a projector, a language backbone or a prediction network, each with a residual width $d$, a modality and a role, and connected by directed edges that form the data path. Each component contains stacks

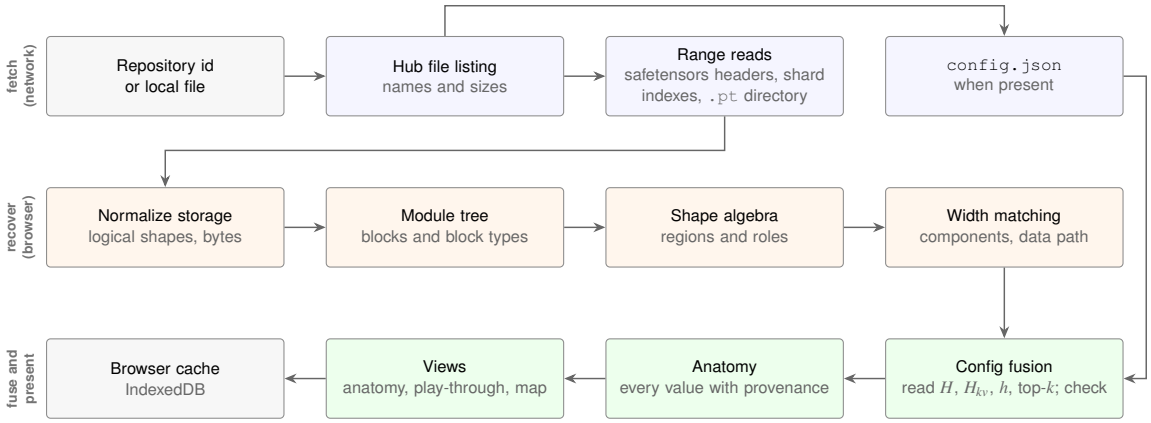



**Figure 1:** Workflow. Only descriptive bytes are fetched: the file listing, safetensors headers, shard indexes, the zip directory and pickle record of PyTorch checkpoints, and the configuration. Recovery and fusion run in the browser, and results are cached locally.

of repeated blocks, and each block is assigned a block type. Each block type is divided into regions, labeled with one of nine roles: attention, cross-attention, feed-forward, mixture of experts, SSM or linear attention, convolution module, adaLN modulation, recurrent cell and norm. Tensors outside blocks are labeled input stem, output head or adapter. Finally, the anatomy records the logical shapes of quantized tensors and several derived quantities: parameters, stored bytes, KV-cache bytes per token and active parameters per token. Every architectural field records its provenance, that is, whether it was measured from the manifest, read from $\mathscr{C}$ and checked against the manifest, or estimated.

### 3.2 What a manifest cannot identify

Several architectural properties leave no trace in tensor shapes. The most important of these is the division of the attention width into heads. A projection $W_q \in \mathbb{R}^{Hh \times d}$ with $H$ heads of size $h$ has the same shape as one with $H/2$ heads of size $2h$, so only the product $Hh$ and the ratio of query to key–value width are visible. The pair becomes identifiable only through per-head parameters, such as query and key norms of width $h$ or relative-position biases of shape $H \times h$. The attention mask is likewise invisible, since a decoder-only language model with tied embeddings stores exactly the same tensors as a bidirectional encoder of the same size. In practice the two usually differ in what surrounds the blocks: BERT-style encoders keep a segment table, a pooler and a norm right after the embedding, and project with biases, while tied decoders end their stack with a final norm and, since Llama [40], omit the biases. These cues are conventions rather than necessities; Table 8 measures how often the family they imply agrees with the configuration. The number of experts is visible but the number used per token is not. Gated and ungated feed-forwards differ in the number of matrices, whereas SiLU and GELU activations cannot be distinguished, and pre-norm and post-norm blocks store identical tensors.

The algorithm therefore reports such fields only as estimates, and the interface labels them accordingly. When a configuration is available, these fields are read from it and checked against the shapes (Section 3.5). The evaluation measures both how often the estimate is correct and how much the configuration adds.

### 3.3 System overview

Figure 1 shows the workflow. Given a repository id, the explorer lists the files in the repository and determines which of them describe the weights. Safetensors files are preferred when they are available. Their shard

**Algorithm 1** Manifest-only anatomy recovery with configuration fusion

**Require:** manifest $\mathscr{M}$, optional configuration $\mathscr{C}$
$\mathscr{T} \leftarrow$ NORMALIZE($\mathscr{M}$) ▷ logical shapes, stored bytes
build the module tree; find lists, expert lists, sub-layer lists, block lists and their owners
**for all** block list $L$ **do**
    group items by structural signature into block types
    **for all** block type $\beta$ **do**
        $d \leftarrow$ width of the leaf norms of a representative block
        classify its children by the rules of Table 1; split nested blocks; pair halves
    **end for**
**end for**
label stems and heads; find adapters, transducer joints, timestep embedders, message passing
choose the backbone and the family; derive $d$, $L$, $f$, $E$, $V$ and KV bytes per token
**if** $\mathscr{C}$ is given **then**
    match sub-configurations to components by width and depth; read $H$, $H_{kv}$, $h$, top-$k$
    check $Hh = x$ and $H_{kv}h = x_{kv}$; mark each field as read from $\mathscr{C}$ or report the mismatch
**end if**
**return** the anatomy, with provenance and evidence for every decision

indexes are read per folder, so a diffusers repository with `transformer/`, `text_encoder/` and `vae/` folders yields one name prefix per component. For each safetensors file, two range requests fetch the length and the header; up to eight files are read in parallel. If a repository contains only PyTorch zip checkpoints, the reader fetches the last 64 KB of the file to find the zip central directory, then fetches only the `data.pkl` record and interprets it with a restricted unpickler that rebuilds names, shapes and dtypes without resolving any Python callable. The configuration file is fetched at the same time. The manifest then passes through the recovery algorithm of Section 3.4 and the result through configuration fusion (Section 3.5). Each step is shown to the user as it happens ("reading header 3 of 12, 41 KB so far"), and the final anatomy is cached in the browser.

## 3.4 Recovery algorithm

The rules in this subsection read only shapes, dtypes and the structure of the name tree. A small lexicon of name fragments (`attn`, `mlp`, `experts` and similar) breaks ties that shapes leave open. This lexicon can be disabled, and Section 5.2 reports results both with and without it. Algorithm 1 summarizes the procedure.

### 3.4.1 Normalizing storage

Quantized checkpoints store one logical weight as several tensors. We group tensors by module and recognize the layout from dtypes and shape ratios before looking at names. NVFP4 stores a `U8` tensor of shape $[o, i/2]$, an FP8 scale tensor $[o, i/16]$ and a scalar second-level scale, so the logical shape is $[o, i]$ and the group size follows from the ratio. FP8 stores an FP8 weight with a scalar, per-channel $[o, 1]$ or block scale. GPTQ stores an `I32` tensor $[i/8, o]$ with scales $[i/g, o]$; AWQ packs the other axis, and the two are told apart by the axis the scales share. MXFP4, MLX, bitsandbytes and compressed-tensors layouts follow the same pattern. Auxiliary tensors count toward stored bytes but not toward parameters.

### 3.4.2 Blocks from the module tree

The dot-separated tensor names form a tree. A node whose children are all integers, or keys such as `expert_0`, `expert_1`, is a list. An item is composite if its subtree holds at least two modules with

| Role | Shape signature inside a block of residual width $d$ |
|---|---|
| Mixture of experts | stacked experts $[E,\cdot,\cdot]$ or an expert list, with a router $[E',d]$, $E \le E' \le E+8$ |
| SSM / linear attn. | depthwise conv $[C,1,k]$, $k \le 8$, with recurrence parameters on the mixer: two or more vectors of one length $n_h \ne d$, or a state tensor with a small last dimension; without a conv, two or more per-head gates $[\le 64,d]$ or decay vectors |
| Feed-forward | one output $[d,f]$ and one or two inputs $[f,d]$, or a fused input $[2f,d]$ with $f > 1.3d$ |
| Attention | one output $[d,x]$ and inputs from $d$, or a fused input $[x+2x_{kv},d]$; low-rank chains that stay inside the region count as internal |
| Cross-attention | as attention, with at least two projections from another component's width $w \ge 128$ |
| Recurrent cell | two or more matrices with $3h$ or $4h$ rows (GRU, LSTM), none returning to $d$ |
| Convolution module | depthwise conv with pointwise $[C,C',1]$ convs or batch-norm statistics |
| adaLN modulation | one matrix $[kd,t]$ from a conditioning width $32 \le t < d$, or $k \in \{6,9,12,18\}$ when $t = d$, or a $[1,k,d]$ table |

**Table 1:** Region signatures. A matrix $[o,i]$ is an input to the region if $i = d$ and an output if $o = d$.

parameters; this separates blocks from `nn.Sequential` wrappers around single layers. An expert list is a list, nested inside an item of another list, whose items are identical two- or three-matrix MLPs without norms. Such a list is usually accompanied by a router whose first dimension equals the number of items. Stacked experts appear instead as 3-D tensors $[E,\cdot,\cdot]$ with one trailing dimension equal to the block width. Block lists are the innermost composite lists that are not expert lists, with one correction for sub-layers: when every item of an outer list holds short inner lists of equal length, as in T5's `block.`*i*`.layer.`*j* or LongCat's two attention and two MLP sub-layers per layer, the inner items are sub-layers and the outer items are the blocks.

Each block list belongs to the module that owns it, moved up through its ancestors as long as no other stack is located there and, in multi-folder repositories, never above its folder. These owners become components, so that embeddings and heads next to an `encoder` wrapper are assigned to the correct component. Within a list, each block gets a signature, the sorted multiset of (relative path with indices masked, tensor rank) pairs, and identical signatures form a block type. In a hybrid stack such as Nemotron-H [30] this gives one type each for the Mamba-2, MoE and attention layers, and the sequence of types is the layer pattern.

### 3.4.3 Regions by shape algebra

For each block type we analyze one representative block. Its residual width $d$ is the length of the vectors held by its leaf norm modules. This estimate is more robust than one based on matrix dimensions, which can be dominated by the inner width of an SSM. The children of the block are then classified with the signatures in Table 1. For attention, vectors in norm-only submodules whose length divides $x$ give the head size, and parameters of shape $[H,h]$ with $Hh = x$ give both factors.

Four refinements are important in practice. A child that holds two sub-modules of different confident roles, such as an attention and an MLP inside one Set Transformer "inducing block" (attention to a few learned points followed by a feed-forward), is split. Two half feed-forwards in sibling modules ($[f,d]$ in `intermediate` and $[d,f]$ in `output`) are paired. If a region is classified with low confidence in the usual layout but is recognized as attention when every matrix is transposed, the weights are stored as $[\text{in},\text{out}]$, as GPT-2's `Conv1D` layers are. Finally, a list whose items map $d$ to another width and never return to $d$ is not a residual stack: its items are read as per-layer prediction heads, such as the box MLPs of DETR-style detectors, or as projectors when the output width belongs to another component, as with the DeepStack mergers of Qwen3-VL, which inject vision features into several decoder layers.

### 3.4.4 Stems, heads and the data path

Outside blocks, a matrix $[V, d]$ with $V > d$ is a token table and a second matrix of the same shape is an untied output head. A vector of length $V$ is the head's bias, and a small table beside a token table or a convolutional stem is a position or type embedding. A chain of matrices whose free input width equals the width $d_X$ of one component, or $k^2 d_X$ for $k \in \{2, 3, 4\}$ after pixel shuffling, and whose free output width equals the width $d_Y$ of another, becomes an adapter with edges $X \to \text{adapter} \to Y$.

Three further patterns are treated as separate components. A transducer has a label embedding $[V, h_p]$ next to a recurrent cell, an encoder projection $[j, d_{\text{enc}}]$ and an output $[V', j]$ with $V \le V' \le V + 16$; the excess $V' - V$ is the number of duration bins of a token-and-duration transducer [43]. A diffusion denoiser has a timestep MLP from a sinusoidal width (128, 256 or 320) to a width $t$ that feeds the modulation matrices in its blocks. A group of square maps $[C, C]$ with an aggregation matrix $[C, rC]$ is a message-passing layer with $r$ reducers. The backbone is the largest component that owns a token table or output head, and the family (decoder, prefix LM, encoder–decoder, transducer, diffusion denoiser, encoder) follows from the components and edges.

## 3.5 Configuration fusion and consistency checking

When a configuration is available, the fields of Section 3.2 are read from it. Configurations of composite models nest several sub-configurations, so we first decide which one describes which component. Every object in $\mathscr{C}$ that declares a width (`hidden_size`, `d_model`, `n_embd`, `dim` and similar) is a candidate. A component is matched to the candidate whose width equals its residual width, preferring one whose layer count equals the component's and whose path indicates the appropriate type of component (`text`, `llm`, `vision`, `audio`). From the match we read the head count, the key–value head count, the head size and the routing top-$k$.

The match is then checked against the weights: the head count times the head size must equal the width $x$ of the attention projections, and the key–value product must equal $x_{kv}$. A field that passes is shown as read from the configuration. A mismatch is reported rather than resolved, because it usually means the field has a different meaning in that family (Section 5.2.4). Without a configuration, the head size falls back to 128 for $d \ge 2048$ and 64 otherwise, and is shown as an estimate.

Parameters and stored bytes are sums over logical and stored tensors. KV-cache bytes per token are $2b \sum_{\ell \in \text{attn}} x_{kv}^{(\ell)}$ for $b$ bytes per element; SSM layers keep a constant state instead. With a top-$k$ value, active parameters per token are the non-expert parameters plus $k/E$ of the experts. The whole procedure makes a few passes over the tree and runs in time close to linear in the number of tensors.

# 4 The explorer

The algorithm is the core of an open web tool, *AI Model Explorer* [38].[1] We first built it for a workshop on edge inference, where participants had to understand a speech model before compressing it, and later generalized it to any architecture. Three goals guided the interface: (1) the first screen should show what the model is made of and how large each part is; (2) each value should state its source, so that a reader can distinguish a measurement from a configuration value or an estimate; and (3) a newcomer should be able to walk through the model from input to output, seeing what each part receives and produces.

The entry screen (Figure 2) offers three entry points and a library of previously analyzed models. The anatomy view (Figure 3) draws the components in data-path order. Each box gives the component's role,

[1] `https://joessattes.github.io/ai-model-explorer/`; the same code is a Python package, `https://pypi.org/project/ai-model-explorer/`.

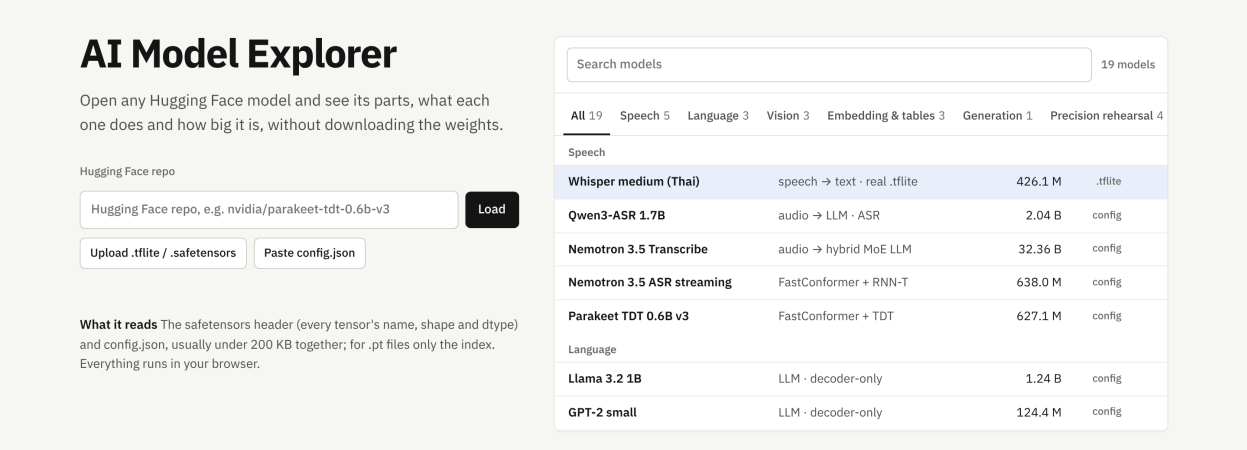



**Figure 2:** Entry screen. A repository id, an uploaded `.safetensors` or `.tflite` file, or a pasted configuration starts an analysis. The library on the right holds analyzed models grouped by task.

parameters, stored size, width and depth, and a strip with one cell per layer colored by block type, so that the layer pattern of a hybrid backbone is visible at a glance. Selecting a component lists its block types with the share of parameters per region and the shape evidence behind each label. A side panel lists the values read from the model, each tagged as measured, read from the configuration or estimated, and states whether the configuration agrees with the weights.

The big-picture view (Figure 4) lays the model out in columns from input to output. Pressing Play steps through the columns in order, and for each column a panel shows what goes in and what comes out, with small stand-ins for the data (a waveform, a spectrogram, a row of tokens, a matrix), the tensor shapes, and one sentence on why the part matters for optimization. The loop at the end follows the family: next token for decoders, next frame for transducers, next denoising step for diffusion models. A layer map draws every layer at its true size under switchable lenses (size, bytes read per sentence, precision), and a report lists where parameters, compute and memory traffic go, together with short observations such as "experts hold 92.8% of the weights; about 13.6 B are active per token". The interface is available in Thai and English.

As an example of use, suppose an engineer wants to run `nvidia/parakeet-tdt-0.6b-v3` on a phone. They paste the repository id and press Load. The explorer lists the repository, fetches `config.json` and the 63 KB header of the 2.5 GB weight file, and completes in less than one second on a typical connection. The anatomy view shows three components: a 24-block conformer encoder with 97% of the parameters, a 12 M prediction network with a two-layer LSTM, and a 6 M joint network whose output size, 8198, is the label vocabulary plus five duration bins. The block view shows that feed-forward modules hold two thirds of each conformer block, which suggests that pruning or low-bit storage would be most effective there. The play-through further shows that the encoder runs once per utterance while the prediction and joint networks run once per emitted label, so streaming latency depends on the small networks rather than on the encoder.

The tool is a single static web page of about 3,000 lines of JavaScript and requires no server. The recovery code is a separate module shared with the evaluation harness, so the numbers in Section 5.2 measure exactly the code that users run.

**Beyond one model.** The same code ships as a Python package with a command-line tool. Besides the anatomy, it answers five practical questions: whether a model fits a given device at a given context length, and with which precision it would (`fits`); what differs between two checkpoints, such as a fine-tune and its base or a quantized copy and the original (`diff`); the anatomy of a list of repositories as one table (`scan`); where a model sits among the popular models of the census (`census`); and the Deploy-Ready verdict of Section 5.5,

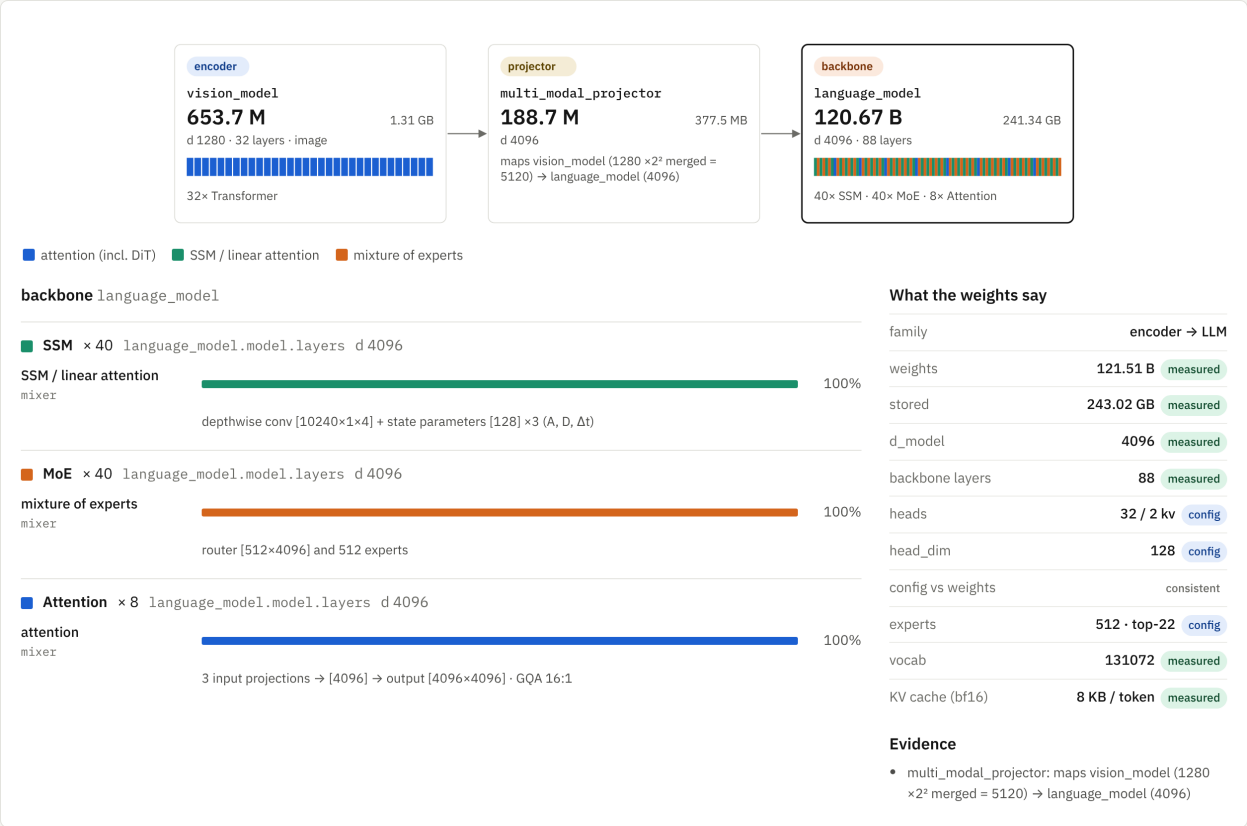



**Figure 3:** Anatomy view of Nemotron 3.5 Super VL (121.5 B). The three components appear in data-path order, and the backbone's strip shows how SSM, MoE and attention layers interleave. Below are the block types with the share of parameters per region and the evidence for each label; on the right, the values read from the model with their provenance and the result of the configuration check.

computed locally (`ready`). Headers of sharded repositories are fetched in parallel and cached per commit. The anatomy of every Hub model links to the corresponding entry on the census page (Section 5.5).

# 5 Experiments

We evaluate the method in four settings of increasing realism: (1) a corpus of 510 architectures whose tensors are labeled from their module classes (Section 5.2); (2) 191 popular checkpoints read from the Hugging Face Hub (Section 5.3); (3) models released in 2026, examined in detail (Section 5.4); and (4) a stratified census of 12,974 Hub repositories (Section 5.5).

## 5.1 Setup

Existing tools either read the safetensors header only for parameter counts [24] or map tensor names to roles through fixed tables for the architectures they support [14]. To our knowledge, neither approach has been evaluated on how much of an architecture the header determines, how robust that determination is when names are uninformative, or how it behaves on checkpoints as they are actually published. We therefore organize the evaluation around ten research questions in three groups. The first five examine recovery on the corpus, where ground truth is available for every tensor:

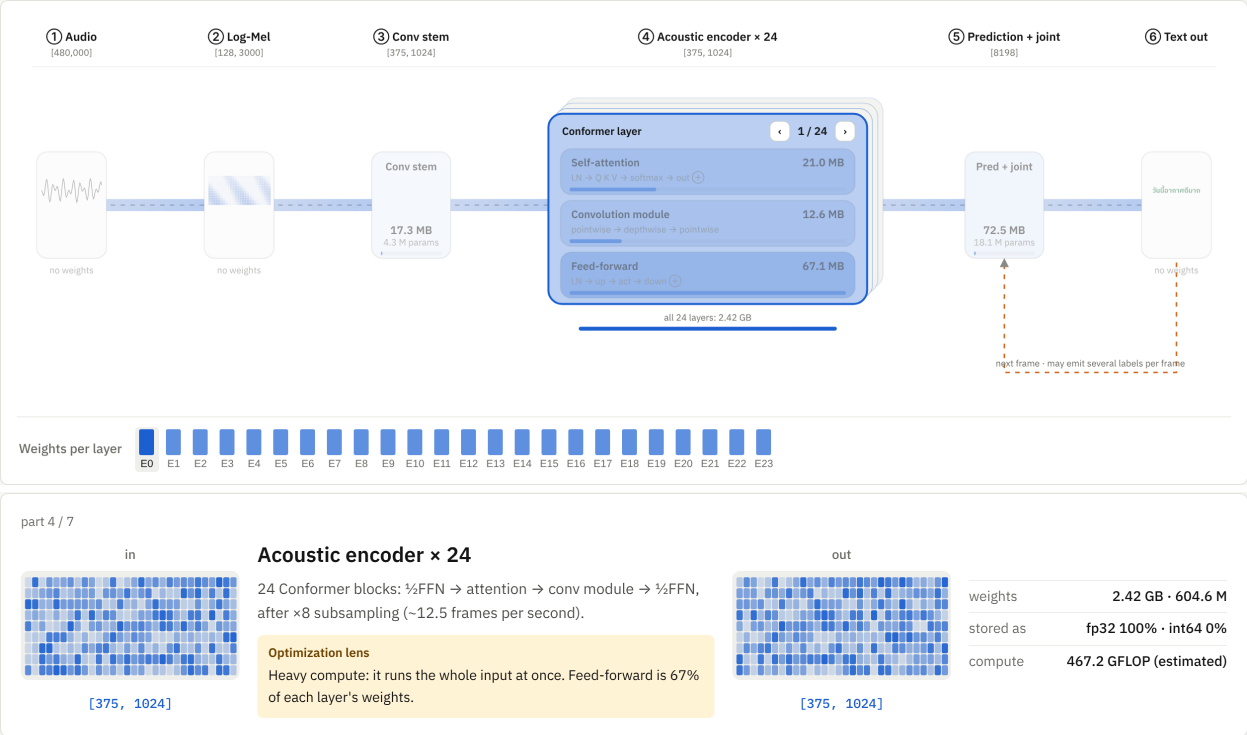



**Figure 4:** Big picture and play-through for Parakeet TDT 0.6B v3. Top: the transducer from audio to text, with the per-frame loop drawn back to the prediction and joint networks. Bottom: step 4 of 7, the acoustic encoder, with its input and output shapes, size, precision and estimated compute.

**RQ1** How accurately are per-tensor roles recovered, and to what extent does accuracy depend on tensor names?

**RQ2** Are stacks, block counts and block types recovered?

**RQ3** Does quantized storage affect the results?

**RQ4** Which hyperparameters can tensor shapes identify, how much does the configuration add, and do configurations agree with the weights they describe?

**RQ5** What does recovery cost in bytes read and in computation?

The sixth asks whether these results transfer beyond default configurations:

**RQ6** Do the results hold on popular checkpoints as they are actually published?

The last four apply the method to the Hub at large, first to assess it at that scale and then to use the recovered architectures to describe the ecosystem:

**RQ7** Does recovery remain accurate across thousands of repositories, and at what cost?

**RQ8** How has the anatomy of released models changed over time?

**RQ9** How do derivative models (fine-tunes, quantized copies and adapters) relate to the models they derive from?

**RQ10** How reliable is the metadata published alongside the weights, and which devices can run each model?

**Corpus.** The `transformers` library, version 5.22.1 [42], registers 788 configuration types. For each, we instantiate the default configuration with the most specific model class available, on PyTorch's meta device,

which creates shapes without allocating memory. Of these, 548 types can be instantiated, and 510 remain after identical manifests are removed. Tied parameters are counted once and persistent buffers are kept, as in a saved checkpoint. The corpus has 268,040 tensors, a median of 419 per model, and spans 1.5 K to 1.74 T parameters with a median of 473 M. Of the 510 models, 475 contain attention, 73 mixtures of experts, 27 SSM or linear-attention mixers, 140 convolutional parts and 70 cross-attention; 231 have more than one stack and 153 have a stack with more than one block type. The configuration of each model is kept for RQ4.

**Ground truth.** The labels are derived from information that the algorithm does not use, namely the Python class of each module. A tensor's role is given by its nearest ancestor whose class name ends in a recognized suffix (`Attention`, `MLP`, `SparseMoeBlock`, `Mamba2Mixer`, `RMSNorm` and so on), with MoE containers overriding their children. Module names are used for labeling in only two cases: a short list of fallbacks for loose tensors, and cross-attention, which is labeled by a path containing `encoder_attn` or `cross_attn` because the library uses the same attention class for both. Tensors that match nothing are labeled *other* and excluded. Convolutional stems and position tables outside block stacks are merged into one input-stem role, because the library is inconsistent about which class owns them. Block lists are module lists whose children each hold at least two modules with parameters, excluding expert containers and using the same sub-layer convention as the algorithm, so RQ2 measures agreement with that convention rather than with an independent definition of a block. Block types are the distinct class compositions of the items. Hyperparameters come from the configuration, using the text sub-configuration for multimodal models.

**Metrics and conditions.** Per-tensor role accuracy is weighted by parameter count within each model and then averaged over models (macro, so that each architecture counts once) or pooled over all tensors (micro, which the largest models dominate). We also report per-role F1 over tensors, recall and precision of block lists by path, the share of matched lists with the exact block count and the exact type partition, exact-match accuracy of hyperparameters, and exactness of the parameter total. The full method uses both names and shapes, whereas the shapes-only condition replaces every non-numeric path token by a consistent random hash and switches the lexicon off, which removes all lexical evidence but keeps the tree; it approximates checkpoints exported by unfamiliar code. The quantized conditions rewrite every 2-D linear weight as NVFP4, FP8 with per-channel scales, or GPTQ 4-bit (Appendix D). We compare against three baselines, all of which read tensor names, as existing tools do. The first is the set of name rules from the previous version of our explorer: regular expressions for layer indices and for fragments such as `q_proj`, `query`, `c_attn`, `mlp`, `fc1`, `encoder_attn` and `mixer`. Its label set has no mixture-of-experts role, so expert tensors are labeled as feed-forward. The second baseline uses the same rules but labels expert, router and shared-expert names as MoE, which removes this disadvantage. The third is the tensor-name table that llama.cpp uses to convert Hugging Face checkpoints [14], the largest maintained mapping of module paths to tensor roles of which we are aware (1,566 paths across its supported architectures, read from `gguf-py/gguf/tensor_mapping.py`). We map each llama.cpp tensor type to our roles, look names up after replacing layer indices, take the union over architectures because the baseline is not told the architecture, and strip expert indices as llama.cpp's converter does. For stacks, all baselines use the usual layer-index heuristic: every `<prefix>.layers.<i>` (also `layer`, `h`, `blocks`, `block`, `resblocks`) is one stack of identical blocks.

## 5.2 Corpus: 510 architectures (RQ1–RQ5)

### 5.2.1 RQ1: Roles

Table 2 summarizes the results; the macro-averaged per-tensor role accuracy is 96.8%. The distribution over models is skewed: the median model is labeled perfectly, 415 of 510 models reach 99% and 473 reach 90%,

| Condition | Role accuracy (%) | | Block lists (%) | | | | Params |
|---|---|---|---|---|---|---|---|
| | macro | micro | recall | precision | count | types | exact |
| Names and shapes | **96.8** | **99.9** | 98.6 | 91.4 | 100 | 92.5 | 510/510 |
| Shapes only | 95.0 | 99.8 | 98.6 | 91.4 | 100 | 92.5 | 510/510 |
| Name rules (v1) | 69.3 | 8.4 | 60.7 | 72.3 | 100 | 83.4 | – |
| Name rules (v1) + MoE label | 82.3 | 98.4 | 60.7 | 72.3 | 100 | 83.4 | – |
| llama.cpp name table | 29.7 | 51.5 | 60.7 | 72.3 | 100 | 83.4 | – |
| NVFP4 storage | 96.8 | 99.9 | 98.6 | 91.4 | 100 | 92.5 | 510/510 |
| FP8 storage | 96.8 | 99.9 | 98.6 | 91.4 | 100 | 92.5 | 510/510 |
| GPTQ storage | 96.8 | 99.9 | 98.6 | 91.4 | 100 | 92.5 | 510/510 |
| Shapes only, NVFP4 storage | 95.0 | 99.8 | 98.6 | 91.4 | 100 | 92.5 | 510/510 |

**Table 2:** Main results on 510 architectures. *Count* is the share of matched lists with the exact number of blocks and *types* the share whose block-type partition equals the ground truth. The baselines share the layer-index stack heuristic, which assumes one block type per list. The micro accuracy of the v1 rules is low because the three largest models, trillion-parameter mixtures of experts, dominate the pooled score and the rules have no MoE label; with that label it rises to 98.4%.

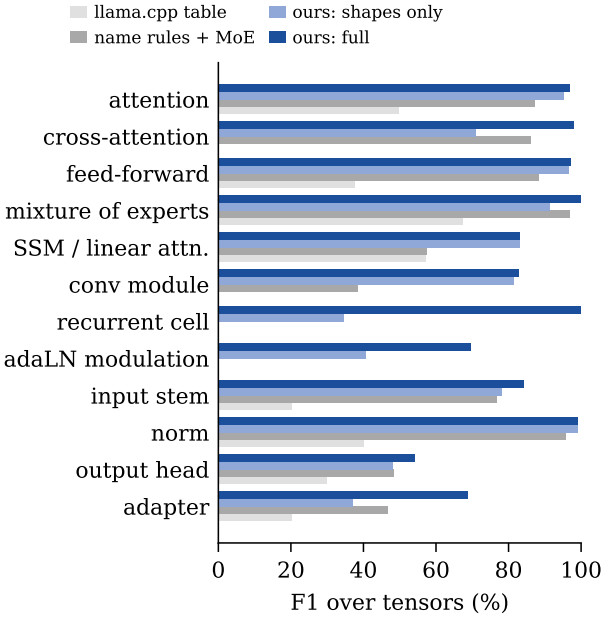



**Figure 5:** Per-role F1 over tensors on the corpus. The two gray bars are the name-reading baselines.

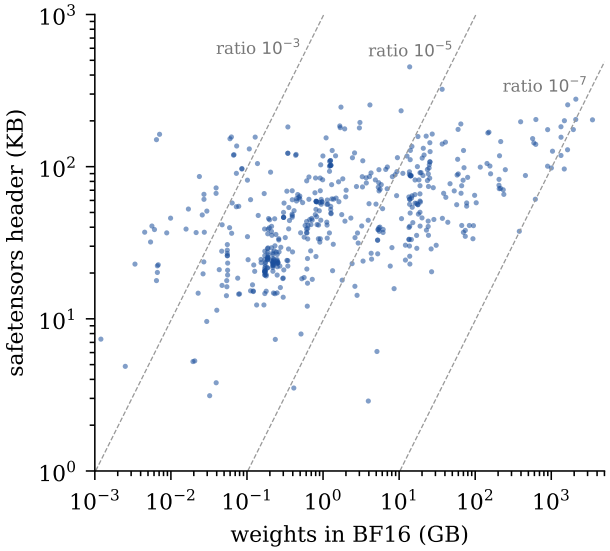



**Figure 6:** Header size against weight size for the 510 manifests. Dashed lines mark constant ratios. The median header is 46 KB and the largest 453 KB.

while 10 fall below 50%. The strongest name baseline, the v1 rules with a MoE label, reaches 82.3%. The llama.cpp table reaches 29.7% overall because it matches only exact module paths: it labels at least 80% of the tensors in 108 of the 510 models, mostly decoder language models. On these 108 models its accuracy is 95.9%, compared with 99.9% for our method. Without names, none of the three baselines can label any tensor, whereas the accuracy of our method decreases by 1.8 points and 374 models still reach 99%. This suggests that recovery relies mainly on the module structure and tensor shapes rather than on vocabulary.

Figure 5 breaks the result down by role. Attention, feed-forward, mixtures of experts and norms exceed 96 F1 with names and stay above 91 without them. Recurrent cells reach 100 with the lexicon but only

35 without it, because an LSTM whose input and hidden widths are equal has exactly the shapes of a feed-forward block that stores a second, integer copy of its weights, as I-BERT does. Cross-attention falls from 98 to 71 without names for a similar reason: when encoder and decoder share a width, the two attention regions of a decoder block have identical shapes, and only a name says which one reads the encoder. Output heads (54) and adapters (69) obtain the lowest scores. Most of the remaining errors arise from poolers and prediction transforms that the class names label as heads but whose shapes resemble a feed-forward layer or an embedding, and from single-matrix projectors inside the language model's folder.

During this study, error analysis led to four changes to the rules, each of which was validated on the whole corpus. Requiring the usual `ih`/`hh` naming before accepting a recurrent cell, when the lexicon is on, removed the I-BERT false positives and raised recurrent-cell F1 from 25 to 100. Treating three equal-width projections next to a convolution as attention fixed ConvBERT's span-based dynamic convolution, and excluding pointwise convolutions from the state-tensor test fixed conformer convolution modules that had been read as SSMs; together these raised convolution F1 from 77 to 83. Reading non-residual lists as heads or projectors raised head F1 from 39 to 54 and adapter F1 from 60 to 69. A rule that would have read every short convolution over the query, key and value channels as linear attention was evaluated and discarded, because it corrected GLM-style gated delta layers but misclassified softmax attention layers that also use short convolutions.

The ten models below 50% are not common architectures; each departs from the block conventions that the rules assume. ALBERT shares one layer across depth and stores no list at all, PoolFormer and PatchTSMixer have blocks without attention or feed-forward signatures, and xLSTM, XLNet, MobileBERT and MobileViTv2 use block layouts that no rule describes. The linear-attention layers of GLM-5-Next keep ordinary query, key and value projections and move the decay parameters into a sub-module, so the shape rules classify them as attention; in EdgeTAM's perceiver the reverse error occurs. For some modules the ground truth itself is debatable. Kimi Delta Attention and the Bailing hybrid store the same tensors as the GLM layers but their classes end in `Attention`, so the class-name oracle calls them attention.

#### 5.2.2 RQ2: Stacks and block types

Of the 1248 ground-truth block lists, 1230 (98.6%) are found at the correct path, and every matched list has the correct number of blocks. The layer-index heuristic finds 60.7% of them, at a precision of 72.3%: it misses lists named differently (`stages`, `conformer_layers`, `transformer_blocks`, `down_blocks`) and cannot see that a list mixes block types. Precision is lower than recall (91.4%) because the algorithm also reports repeated sub-blocks inside convolutional backbones, such as ResNet bottleneck sub-layers and the residual units of the SEANet audio codec, which the ground-truth convention does not count. The block-type partition is exact for 92.5% of matched lists. Most of the misses are lists whose first block differs only in a module that is reflected in the class signature but not in the tensor signature, such as a relative-position bias owned by layer 0.

#### 5.2.3 RQ3: Quantized storage

Under NVFP4, FP8 and GPTQ storage the results are identical to those at full precision, and parameter totals stay exact for all 510 models (Table 2). This also holds with names hashed, because packed weights and their scales are recognized by dtype and shape ratios rather than by suffixes such as `weight_scale`. This is relevant in practice because the Hub counts stored elements: an NVFP4 checkpoint, which stores two values per byte, is listed at a little over half of its parameter count, and a bitsandbytes 4-bit checkpoint slightly above it (Section 5.3).

| | $d$ | layers | FFN | experts | vocab | heads | KV heads | head size |
|---|---|---|---|---|---|---|---|---|
| Names and shapes, all models | 92.7 | 91.8 | 90.5 | 97.3 | 87.7 | 73.0 | 69.9 | 66.4 |
| number of models | 467 | 441 | 336 | 73 | 359 | 445 | 226 | 152 |
| Names and shapes, single stack | 94.7 | 96.3 | 93.5 | 96.4 | 85.6 | 70.4 | 63.9 | 57.1 |
| Shapes only (names hashed) | 92.7 | 91.8 | 90.5 | 94.5 | 85.0 | 73.0 | 69.9 | 66.4 |
| With configuration fusion | – | – | – | – | – | 95.7 | 98.2 | 96.7 |
| Identifiable from shapes | yes | yes | yes | yes | yes | only with per-head parameters | | |

**Table 3:** Exact-match accuracy (%) of hyperparameters against the configuration. FFN is evaluated on dense models only.

#### 5.2.4 RQ4: Identifiability, configuration fusion and consistency

Table 3 separates the fields shapes can identify from those they cannot. Width, depth, FFN size and expert count are correct for 90 to 97% of models; width, depth and FFN size are correct more often when a model has a single stack, because then there is no doubt about which component the configuration describes. From shapes alone, the head count is correct for 73% of models and the head size for 66% of the 152 whose configuration declares it. When the head size behind the count comes from the default (128 for $d \geq 2048$, 64 otherwise), the count is correct for 78% of models; when it comes from a norm vector of plausible length, it is correct for 72%. For 20 models, no head size is consistent with the projection widths, and no count is reported. In latent attention, the norm before the low-rank key–value projection has the latent width rather than the head width, which is a concrete case of the limit described in Section 3.2.

With configuration fusion, head count, key–value head count and head size are correct for 95.7%, 98.2% and 96.7% of models. Since the ground truth comes from the same configuration, these numbers measure the matching step, that is, whether the correct sub-configuration is attached to the backbone. The backbone found a match in 455 of 510 models, and 872 of 1106 components found one. Only the consistency check is independent of the configuration. Of the 636 matched components with attention, 605 (95.1%) have configurations whose head fields multiply out to the projection widths in the weights. The 31 disagreements are concentrated in latent-attention models (DeepSeek-V2, V3 and V4, Kimi, MiniCPM3, Mistral 4, LongCat-Flash), where `head_dim` or `num_key_value_heads` refers to the query–key or latent geometry rather than to a projection width, in Falcon's fused multi-query layout, and in SAM-style windowed vision encoders. We do not consider these configurations incorrect; each contains a field whose meaning is specific to the model family. A tool that relied on the configuration without checking it would, however, report incorrect KV-cache sizes for these models, and the explorer therefore displays the disagreement instead of resolving it.

#### 5.2.5 RQ5: Cost

The median safetensors header is 46 KB, the 95th percentile 164 KB and the largest 453 KB, for models of up to 3.5 TB in BF16 (Figure 6). Recovery takes 10 ms for the median model on a single JavaScript thread, 41 ms at the 95th percentile and 98 ms at most. In the explorer, the cost is dominated by network round trips: a single-file model needs four requests (listing, configuration and two range requests), and a sharded model two range requests per shard, issued in parallel.

### 5.3 Popular checkpoints on the Hub (RQ6)

The corpus measures accuracy against independent labels, but its manifests come from one library's default configurations. To test the method on checkpoints as they are actually published, we selected 201 popular public repositories across twelve groups: dense and mixture-of-experts language models, state-space and

| Repository | Storage | Ours | Hub shows | Header | Recovered structure |
|---|---|---|---|---|---|
| deepseek-ai/DeepSeek-V3 | FP8 | 684.5 B | 684.5 B | 11.3 MB | 61 layers, 256 experts, + MTP layer |
| nvidia/DeepSeek-R1-0528-FP4 | NVFP4 | 684.5 B | 396.8 B | 20.4 MB | same; the Hub counts packed bytes |
| moonshotai/Kimi-K2-Instruct | FP8 | 1.03 T | 1.03 T | 16.9 MB | 61 layers, 384 experts |
| zai-org/GLM-4.5 | BF16 | 358.3 B | 358.3 B | 5.3 MB | 92 layers, 160 experts + MTP layer |
| Qwen/Qwen3-235B-A22B | BF16 | 235.1 B | 235.1 B | 4.4 MB | 94 layers, 128 experts, 64/4 heads |
| MiniMaxAI/MiniMax-M2 | FP8 | 228.7 B | 228.7 B | 12.0 MB | 62 layers, 256 experts |
| openai/gpt-oss-120b | MXFP4 | 116.8 B | 116.8 B | 76 KB | 36 layers, 128 experts, 64/8 heads |
| Qwen/Qwen3-Next-80B-A3B-Instruct | BF16 | 81.3 B | 81.3 B | 9.0 MB | 48 layers, linear + full attn., 512 exp. |
| Qwen/Qwen2.5-VL-72B-Instruct | BF16 | 73.4 B | 73.4 B | 151 KB | vision 32 L $\rightarrow$ merger $\rightarrow$ 80 L |
| unsloth/Llama-3.3-70B-Instruct | BF16 | 70.6 B | 70.6 B | 83 KB | 80 layers, 64/8 heads, FFN 28672 |
| mistralai/Mixtral-8x7B-v0.1 | BF16 | 46.7 B | 46.7 B | 125 KB | 32 layers, 8 experts |
| unsloth/gemma-3-27b-it | BF16 | 27.4 B | 27.4 B | 163 KB | vision 27 L $\rightarrow$ projector $\rightarrow$ 62 L |
| nvidia/NVIDIA-Nemotron-Nano-9B-v2 | BF16 | 8.89 B | 8.89 B | 37 KB | 56 layers: Mamba-2, FFN, attention |
| unsloth/Qwen3-8B-bnb-4bit | NF4 | 8.19 B | 8.41 B | 193 KB | 36 layers, shapes rebuilt |

**Table 4:** A selection of the 191 checkpoints read from the Hub. *Ours* is the logical parameter count; *Hub shows* is the count on the model page. *Header* is the number of bytes read. MTP: an extra multi-token-prediction layer used for speculative decoding.

| | $d$ | layers | FFN | experts | vocab | heads | KV heads | head size |
|---|---|---|---|---|---|---|---|---|
| Number of models | 155 | 154 | 112 | 31 | 145 | 149 | 94 | 56 |
| Names and shapes | 95.5 | 92.9 | 90.2 | 100 | 93.1 | 77.2 | 75.5 | 83.9 |
| With configuration fusion | – | – | – | – | – | 96.6 | 97.9 | 96.4 |

**Table 5:** Exact-match accuracy (%) of hyperparameters on the Hub checkpoints, against each repository's `config.json`.

hybrid models, encoders, embedding models, sequence-to-sequence models, speech and audio models, vision backbones, vision–language models, image and video generators, time-series and tabular models, and quantized releases. For gated models, we used ungated mirrors where one existed. Together the repositories hold 12.1 TB of weights and were downloaded about 505 million times in the month before the run. A benchmark page in the explorer (`bench.html`) runs, in the user's browser, the same loader as the explorer and exports each manifest, the configuration files and the Hub's metadata; the analysis then runs offline with the same recovery code.

Of the 201 repositories, 191 were read successfully. Of the ten failures, seven are gated and two store old non-zip pickles. The remaining one, MMS-1B-all, ships one adapter file per language, and reading all of their headers triggered the Hub's rate limit; the loader now skips such adapters. Reading the 191 manifests transferred 161 MB in total. The median repository needed 79 KB, the 95th percentile 5.3 MB, and the largest, an NVFP4 release of DeepSeek-R1 with 163 shards and 180,488 tensors, 20.4 MB. Recovery took 18 ms for the median model and 4.2 s for the largest.

**Parameter counts.** The Hub's model pages show a parameter count computed from safetensors metadata, which gives an independent check of the header reading. Excluding 0-d scalars such as batch-norm step counters, which the Hub skips, our count equals the Hub's for 149 of the 158 transformers-style repositories that have one: the stored count for full-precision files, and the logical count for FP8, GPTQ, AWQ, MXFP4 and MLX releases, for which the Hub reports logical sizes. For ten of the 149 the two agree only once the 0-d scalars are excluded. For the 12 diffusers pipelines with a count, the Hub's figure exactly equals the number of parameters we assign to the denoiser (the UNet or transformer folder), while our total also includes the text encoders and VAE. Of the nine differences, five are 4-bit releases where the Hub counts packed storage: an NVFP4 DeepSeek-R1 is shown at 396.8 B instead of 684.5 B (Table 4), two NVFP4 8 B models at about

| Group | Repos | Read | Hub = | Heads (fused) | Roles | Median header | Median time |
|---|---|---|---|---|---|---|---|
| Dense LM | 48 | 45 | 44/44 | 45/45 | 99.7% (32) | 38 KB | 10 ms |
| MoE LM | 24 | 24 | 23/24 | 24/24 | 100.0% (3) | 2,229 KB | 685 ms |
| SSM / hybrid | 10 | 9 | 8/8 | 4/6 | 60.4% (4) | 66 KB | 15 ms |
| Encoder | 8 | 7 | 6/6 | 6/7 | 90.8% (6) | 23 KB | 5 ms |
| Embedding | 10 | 10 | 9/9 | 10/10 | 100.0% (1) | 34 KB | 12 ms |
| Seq2seq | 7 | 6 | 3/3 | 6/6 | 100.0% (6) | 155 KB | 13 ms |
| Speech / audio | 25 | 23 | 15/16 | 17/17 | 99.7% (11) | 123 KB | 16 ms |
| Vision | 19 | 19 | 16/16 | 7/9 | 94.7% (6) | 62 KB | 20 ms |
| Vision–language | 19 | 18 | 15/17 | 12/12 | – | 107 KB | 31 ms |
| Image / video gen. | 15 | 14 | 0/0 | – | – | 338 KB | 83 ms |
| Time series / tabular | 6 | 6 | 5/5 | 3/3 | 97.7% (2) | 31 KB | 8 ms |
| Quantized release | 10 | 10 | 5/10 | 10/10 | 100.0% (1) | 109 KB | 20 ms |

**Table 6:** Hub benchmark by group. *Hub* = counts transformers-style repositories whose total agrees with the model page; diffusers pipelines are excluded here (their Hub count equals our denoiser in all 12 cases). *Heads (fused)* is the head count after configuration fusion, against `config.json`; *Roles* is macro role accuracy on models whose type is in the corpus (number of models in brackets).

57% of their size, and two bitsandbytes releases 2 to 3% above it, because packed bytes, block scales and quantization maps are counted as parameters. Two come from files that the Hub does not count, a codec stored beside a speech model and the LoRA folders shipped with Phi-4-multimodal; in one FP8 release of DeepSeek-V3.1 the Hub also counts the 41.5 M block scales; and one robotics policy differs by 72 elements whose origin we could not identify.

**Hyperparameters and roles.** Table 5 repeats the analysis of RQ4 on the Hub checkpoints. The pattern is the same as on the corpus, with higher numbers because popular checkpoints are mostly single-stack models with standard layouts. From the manifest alone, the head count is correct for 77% of models; with fusion it is correct for 97%, and the configuration agrees with the weights for 136 of the 137 checked models (the exception is Phi-4-multimodal). For 72 checkpoints whose model type is also in the corpus, we transferred the class-name labels by index-masked tensor name. Role accuracy on these is 96.3% macro, and 61 of the 72 reach 99%, against 90.8% for the v1 rules with a MoE label and 59.5% for the llama.cpp table. On the 42 of these checkpoints that the llama.cpp table covers well (at least 80% of tensors labeled), it reaches 97.3% and our method 99.7%. For the layer count, the layer-index heuristic matches the configuration for 138 of 154 models and our method for 143. Four checkpoints have role accuracy below 90%: xLSTM and ALBERT, which also fail on the corpus, Zamba2, and RT-DETR, whose CNN backbone the rules split into many small stacks. Table 6 breaks the results down by group, and Appendix F lists every checkpoint.

**What real checkpoints added.** Four problems appeared only on real files, and each led to a change that was then checked on the corpus. DeepSeek-V3 and GLM-4.5 store their multi-token-prediction layer as one extra item of the main layer list. It is now recognized by its $[d, 2d]$ projection and two extra norms and reported separately, so the layer count matches the configuration (61 and 92 rather than 62 and 93). bitsandbytes releases keep the logical shape in a serialized state blob that the header does not expose; the loader now regroups the state tensors with their weight, counts two values per packed byte, and rebuilds a matrix shape from the residual width, which recovers the 8.19 B of Qwen3-8B. In Qwen3-0.6B the key and value projections are square $[d, d]$, which had caused one of them to be mistaken for the output projection; the output is now taken from the non-square side. Whisper's encoder stores a position table with 1,500 rows that had been read as a token table, which made the encoder look like the backbone. A component fed by a

| Model | Family | Params | Header | Recovered structure |
|---|---|---|---|---|
| Nemotron 3.5 Super VL | prefix LM | 121.5 B | 130 KB | ViT 32 L → projector → 40 SSM, 40 MoE, 8 attn. |
| Nemotron 3.5 Transcribe | prefix LM | 32.4 B | 154 KB | 3 encoders → projector → 23 SSM, 23 MoE, 6 attn. |
| Nemotron 3.5 ASR streaming | transducer | 638 M | 57 KB | 24 conformer blocks, LSTM, RNN-T joint |
| Parakeet TDT 0.6B v3 | transducer | 627 M | 63 KB | same encoder, TDT joint, 5 duration bins |
| Nemotron 3 Embed 1B NVFP4 | encoder | 1.14 B | 41 KB | 16 blocks, 112 NVFP4 linear layers |
| FastH3 Omni FP8[†] | diffusion | 33.1 B | 84 KB | 2 refiner + 50 DiT blocks, 300 FP8 tensors |
| Kumo Tabular (large)[†] | encoder | 214 M | 41 KB | row embedding → 24-layer backbone |
| Kumo Relational[†] | encoder | 29 M | 24 KB | rows → message passing → 12 layers |

**Table 7:** Anatomies recovered for models released in 2026. No manifest here comes from the published weight files: each was built from the model's published configuration through its `transformers` implementation (the NVFP4 embedding model then stored in the published NVFP4 layout), or, marked [†], rebuilt from public source code because `transformers` has no implementation.

convolutional stem is no longer treated as having a token table. On the loading side, repositories that ship several copies of the same weights (FP16 and FP32 files, single-file checkpoints beside a diffusers layout, LoRA files) are now reduced to one description, because the first run read some of them more than once.

## 5.4 Models released in 2026

Table 7 lists eight models released in 2026 that we studied in detail. Their model types are also in the corpus, with the library's default configurations; here each is built from its own published configuration by instantiating its `transformers` implementation without weights, so sizes and layouts are those of the released models. They were not held out from development: we studied them while extending the rules, and several parts of Section 3.4, including transducer joints, timestep embedders, message-passing layers and adapter components, were added in response to them. Every change was checked on the full corpus for regressions. The case studies therefore indicate that the representation is expressive enough to describe these architectures; accuracy is estimated on the corpus (Section 5.2), with the caveat discussed in Section 6, and on the published checkpoints of Section 5.3.

The Super VL backbone mixes three block types in one list, and the recovered type sequence reproduces the configuration's `layers_block_type` exactly. The projector's input width, 5120, is four times the vision width, and the algorithm reports a $2 \times 2$ merge, matching the configuration's `downsample_ratio` of 0.5. Experts hold 112.7 B of the 121.5 B parameters. With the configured top-22 routing this gives 13.6 B active parameters per token, including the vision tower and both embedding tables, compared with the 12 B indicated by the model's name. Only 8 of the 88 layers keep a KV cache, each with two key–value heads, so the cache costs 8 KB per token in BF16, an eleventh of what it would be if every layer used the same attention.

For both speech transducers the joint network is found from three shapes, and the number of duration bins follows from the difference between two of them. Parakeet's joint output has 8198 rows and its label table 8193, giving the five durations $\{0, \ldots, 4\}$ of its configuration, while the streaming model's two sizes are equal, as in a standard RNN-T. Parakeet's parameter total exceeds the Hub's figure by 24, its batch-norm step counters: one 0-d scalar per conformer block, which the Hub skips (Section 5.3). The NVFP4 embedding model has 1.14 B logical parameters, matching its model card, while the Hub reports 0.7 B because it counts packed `U8` elements. The weights alone do not reveal that this model is an encoder; the setting `is_causal: false` in its configuration resolves this, and the explorer therefore reports the family given by the configuration.

Besides the new components, four fixes to existing rules came from these models: the residual width of a block is taken from its norms, `nn.Sequential` children are no longer split as sub-layers, a module that

wraps two different sub-blocks is split, and in repositories with several top-level folders a component is never raised above its folder. One component remains unexplained. The language-prompt projector of the streaming model is not attached to any other component, because its prompt table lives outside the components the manifest reveals.

### 5.5 A census of the Hugging Face Hub (RQ7–RQ10)

The checkpoints of Section 5.3 were selected manually. To assess how the method performs on the Hub as a whole, and what the weights reveal about the Hub itself, we applied it to a stratified sample of model repositories. Architectural properties below come from the tensor shapes, with head geometry and routing taken from configurations checked against them (Section 3.5). Everything else comes from Hub metadata: the relation to a parent model, the pipeline, the license, creation dates and download counts.

**Summary of findings.** (1) Headers totaling 4.9 GB describe 283 TB of weights, about 58,000 bytes of weights per byte of header. (2) Among popular original language models, mixtures of experts went from 0% of those created up to 2022 to 42% of 2026 releases, grouped-query attention from 0% to 84%, and the KV cache per token per billion parameters fell 24-fold; all trends remain significant when the first year bin is excluded. (3) Mixtures of experts became sparser: from 8 experts with 28% of parameters active per token in 2024 to 256 experts with 11% in 2026. (4) As released dense models grow larger, their width increases faster than their depth, $d \propto N^{0.40}$ against $L \propto N^{0.28}$. (5) Quantized copies are 6% of repositories but receive 28% of downloads, and 73% of the fine-tunes whose parent was also read keep the parent's architecture exactly. (6) The Hub's parameter count differs from the logical count for 12.2% of readable popular repositories and is missing for 24.4%, and 13% of popular models above 1B parameters store FP32. (7) 96% of original language models created up to 2022 fit a 16 GB laptop at 4-bit with an 8k context, against 62% of 2026 releases.

**Statistics.** The popular strata are read in full. We treat them as one realization of the release process and give 95% intervals for that process, not for sampling error: Wilson intervals [41] for shares, bootstrap intervals [9] for medians, and, for scaling slopes, a bootstrap that resamples providers, since checkpoints from one provider share their design. Trends over creation year are tested with the Cochran–Armitage test [2, 5] with the year as score and a logistic odds ratio per year; because the first bin is open-ended, we also report the odds ratio fitted on 2023–2026 alone. Monotone trends in continuous quantities are tested with Spearman's $\rho$, and differences between two groups with the Mann–Whitney test [27]. Proportions for the Hub as a whole use the sampling weights. A *language model* here is a repository above 50M parameters whose family is decoder or prefix LM (a decoder that reads the output of an image or audio encoder as a prefix, Appendix A) and whose Hub pipeline generates text; repositories without a pipeline count only if their configuration also describes a generative decoder. The family is the one recovered from the weights, or the configuration's when the two disagree: about 4% of generative language models, mostly small decoders with tied embeddings, are recovered as encoders, and leaving them out would bias the results for small models. Without these conditions, dual encoders such as CLIP and encoder–decoders such as T5 that the family rules place among decoders would make up a large part of the earliest year. The Hub dates every repository created before March 2022 to that month (7% of popular repositories carry this date), so the first bin reads *up to 2022* and contains only models that are still popular now. All analyses are in `ecosystem.py` and run on the released data.

| | Originals ($n$=3,506) | | Frontier ($n$=494) | | 2026 releases ($n$=1,751) | |
|---|---|---|---|---|---|---|
| | manifest | fused | manifest | fused | manifest | fused |
| $d$ | 97.3 | | 98.1 | | 96.1 | |
| layers | 94.6 | | 88.9 | | 92.1 | |
| heads | 81.0 | 97.3 | 75.2 | 97.4 | 78.9 | 96.1 |
| KV heads | 75.9 | 98.8 | 69.2 | 98.4 | 74.9 | 98.1 |
| head size | 80.4 | 98.0 | 76.1 | 98.2 | 82.3 | 97.7 |
| FFN | 95.7 | | 95.6 | | 92.3 | |
| vocab | 93.0 | | 92.6 | | 92.3 | |
| experts | 93.8 | | 97.0 | | 91.8 | |
| family | 95.9 | | 92.9 | | 94.9 | |

**Table 8:** Hyperparameters recovered from the weights of published checkpoints, against the values their `config.json` declares (exact match, %). *Fused*: head geometry after configuration fusion. FFN on dense models only. Test fixtures and placeholder configurations are excluded.

### 5.5.1 RQ7: Recovery at Hub scale and its cost

**Sampling.** The frame is the Hub's daily metadata dump [11]:[2] 3.2 M public model repositories with their download and like counts, creation dates, libraries and tags. The 10,000 most downloaded repositories receive 95% of all downloads in a month, and 44% of repositories were not downloaded in that month. We read in full the 10,000 most downloaded repositories, which already include more than 3,000 *original* releases (those the Hub does not tag as a quantized, fine-tuned, merged or adapter derivative), and 743 original releases created since 2025-11-01 by fifty leading providers that we listed before the run (the list is in the census code), at most forty per provider, of which 470 were not among the top 10,000. These strata represent the most widely used models; after removing repositories that exist only to test software (tiny random weights downloaded millions of times by continuous integration), 9,939 remain. The 12,974 repositories read include 552 such test repositories, which are left out of every analysis. A further 2,504 repositories are drawn at random from the rest of the Hub, stratified by creation year, download tier and the presence of safetensors, with weights equal to stratum size over sample size, so that proportions can be stated for the Hub as a whole.

**Reading.** Each repository is read as in the explorer: the file list, `config.json`, and two range requests per safetensors file, with all shards fetched in parallel, in a separate process with a 120-second limit, within the Hub's rate limits. 12,974 repositories were read in about three and a half hours on one machine. For the median readable repository, reading required 80 KB of headers (0.0021% of its weight bytes) and took 1.8 s. Of the popular repositories, 67% were readable, 27% hold no safetensors or PyTorch weights, 2.0% are gated, 2.8% store pre-zip PyTorch pickles, and 1.3% timed out or failed. Of the repositories with no weight files, 87% ship only GGUF files, a format that the census does not yet support. The raw manifests and configurations are kept, so every number below can be recomputed offline after a rule change; they are released with the census.[3] The recovery rules were frozen at public commit `a0afb09`; all census results were computed with that commit, by re-scoring the stored manifests.

**Accuracy in the wild.** Table 8 scores the recovered hyperparameters against the values each repository's configuration declares, on 3,506 original releases, 494 recent releases by the leading providers (the frontier

[2] https://huggingface.co/datasets/cfahlgren1/hub-stats, snapshot of 2 November 2026.

[3] Dataset: https://huggingface.co/datasets/JoesSattes/hub-weight-census; interactive findings and a searchable table of models: https://joessattes.github.io/ai-model-explorer/census/; code: https://github.com/JoesSattes/ai-model-explorer/tree/main/research/hub-census.

| Popular original LMs | ≤2022 ($n$=46) | 2026 ($n$=425) | OR / year | OR / year, 2023–26 | trend |
|---|---|---|---|---|---|
| Mixture of experts | 0% (0–8) | 42% (38–47) | 2.73 (2.30–3.23) | 2.69 (2.26–3.19) | $p < 10^{-10}$ |
| Grouped-query attention | 0% (0–8) | 84% (81–88) | 2.00 (1.78–2.26) | 1.71 (1.50–1.95) | $p < 10^{-10}$ |
| Hybrid (state-space / linear attn.) | 0% (0–8) | 28% (24–33) | 10.11 (6.08–16.83) | 10.11 (6.07–16.83) | $p < 10^{-10}$ |
| Latent attention | 0% (0–8) | 8% (6–11) | 3.32 (2.06–5.34) | 3.29 (2.03–5.33) | $p < 10^{-7}$ |
| Vision/audio-language (prefix LM) | 0% (0–8) | 44% (40–49) | 2.55 (2.19–2.99) | 2.50 (2.14–2.94) | $p < 10^{-10}$ |
| Published already quantized | 0% (0–8) | 24% (20–28) | 2.40 (1.95–2.96) | 2.36 (1.90–2.93) | $p < 10^{-10}$ |
| 32k-token KV cache > 4-bit weights | 93% (82–98) | 34% (30–39) | 0.56 (0.51–0.63) | 0.59 (0.53–0.67) | $p < 10^{-10}$ |
| Fits a 16 GB laptop at 4-bit, 8k | 96% (85–99) | 62% (57–66) | 0.76 (0.68–0.85) | 0.81 (0.71–0.91) | $p < 10^{-5}$ |
| Weights stored in FP32[†] | 95% (93–96) | 30% (27–33) | 0.45 (0.43–0.48) | 0.56 (0.52–0.60) | $p < 10^{-10}$ |

**Table 9:** How popular original language models changed by creation year. Anatomy from the weights (head geometry checked against the configuration); origin, pipeline and creation year from Hub metadata. Shares with Wilson 95% intervals; odds ratio per year from a logistic fit (95% Wald interval), over all years and over 2023–2026 alone; trend $p$ from the Cochran–Armitage test over all five year bins. [†]All popular original releases, not only language models ($n$=852 and 902). The Hub dates repositories created before March 2022 to that month, hence the first bin.

stratum, readable and with a configuration) and 1,751 repositories created in 2026, across 658 model types from 1,734 providers. The pattern observed in Table 3 also holds at this scale: width, depth and expert count are read from the weights alone, and head geometry needs the configuration, after which it is correct for 96 to 99% of models. The family agrees with the declared architecture class for 95.9% of original releases and 94.9% of 2026 releases, which also measures how often the encoder and decoder cues of Section 3.2 hold. The frontier set is not a clean holdout, since the rules were last revised with earlier census data in view; models created after the freeze will provide a prospective test, which we intend to report as more such models become available.

**Roles on real checkpoints.** The corpus provides per-tensor labels for every model type it contains, and the census manifests allow us to transfer these labels to published checkpoints. For each census repository whose configuration names a model type in the corpus, we label its tensors by index-masked name with the class-name roles of that type, and score repositories where at least half the tensors find a label. On 2,269 repositories of 167 model types, macro role accuracy is 98.7%, and 92% of repositories reach 99%; the name rules with a MoE label reach 95.3% and the llama.cpp table 33.2%. These are published files with their own storage and quantization; because labels are matched by name, the test covers checkpoints that follow their model type's naming, and it complements the corpus result rather than repeating it.

### 5.5.2 RQ8: How the anatomy of released models changed

**Experts, grouped attention and multimodal input.** Table 9 and Figure 7 follow 1,251 popular original language models by creation year. Mixtures of experts went from 0% (0–8) to 42% (38–47) (odds ratio 2.73 (2.30–3.23) per year). Grouped-query attention, in which several query heads share one key–value head, went from 0% (0–8) to 84% (81–88), with a median of 4 query heads per key–value head among 2025–2026 releases. Hybrid stacks, which replace some attention layers with state-space or linear-attention layers, stayed below 5% until 2025 and reached 28% (24–33) in 2026; latent attention, which caches a compressed key–value projection, reached 8% (6–11). Prefix LMs, almost all of them vision- or audio-language models, went from 0% (0–8) to 44% (40–49). The median vocabulary grew from 50,257 to 151,936 tokens and the median head size from 64 to 128, counting only models whose head size is observed (from the configuration or a query–key norm) rather than set by the default of Section 3.5. The share of models released in quantized form rose from 0% (0–8) to 24% (20–28).

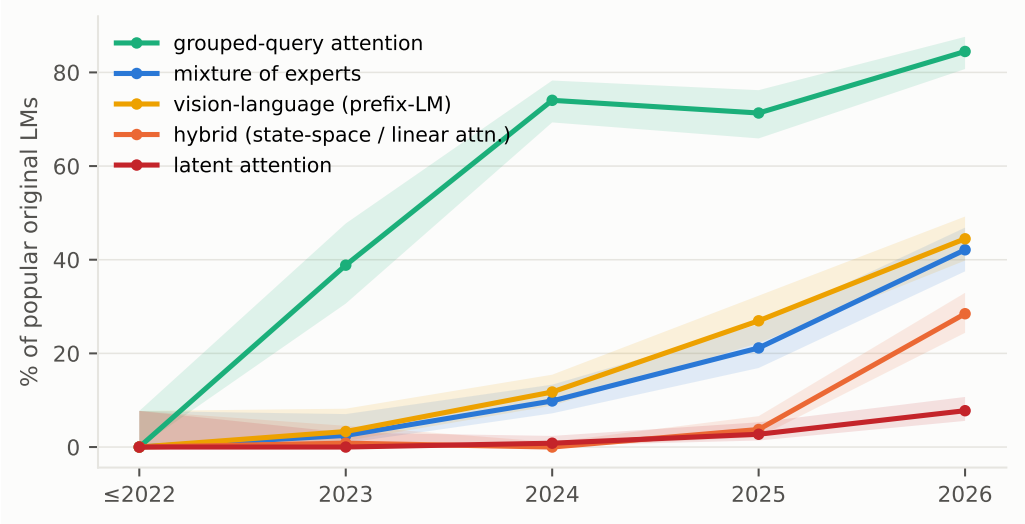



**Figure 7:** Popular original language models by creation year, with Wilson 95% intervals (shaded). Every trend is significant (Table 9).

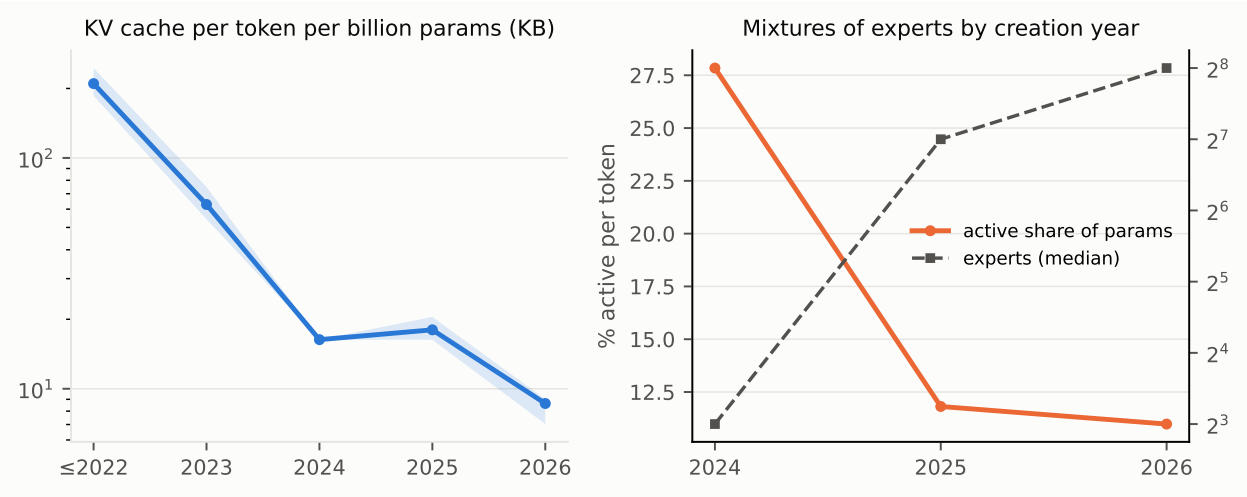



**Figure 8:** Left: median KV cache per token per billion parameters (16-bit cache, log scale, bootstrap 95% interval). Right: mixtures of experts in years with at least 20 of them, median share of parameters active per token and median number of experts per layer (log scale).

**Memory per token.** Once the weights are loaded, the size of the KV cache determines how much context fits in memory. Normalized by model size, it fell from a median of 210 KB per token per billion parameters up to 2022 to 8.6 KB in 2026 (95% interval 7.0–9.0 KB), a 24-fold drop (Spearman $\rho = -0.38$ with creation year, $p < 10^{-10}$; Figure 8), although the decline is not monotone, since 2025 releases lie slightly above those of 2024. At 32k tokens a 16-bit cache outweighed 4-bit weights for 93% (82–98) of models up to 2022 and for 34% (30–39) of 2026 releases (Cochran–Armitage $z = -10.6$, $p < 10^{-10}$); models without a KV cache are counted as not exceeding the weight size.

**Sparser mixtures of experts.** Among original mixtures of experts, the median number of experts per layer went from 8 in 2024 ($n = 36$) to 256 in 2026 ($n = 179$). The number routed per token and the active share need the configuration's top-$k$, which is missing for some models; on those that declare it, the median top-$k$

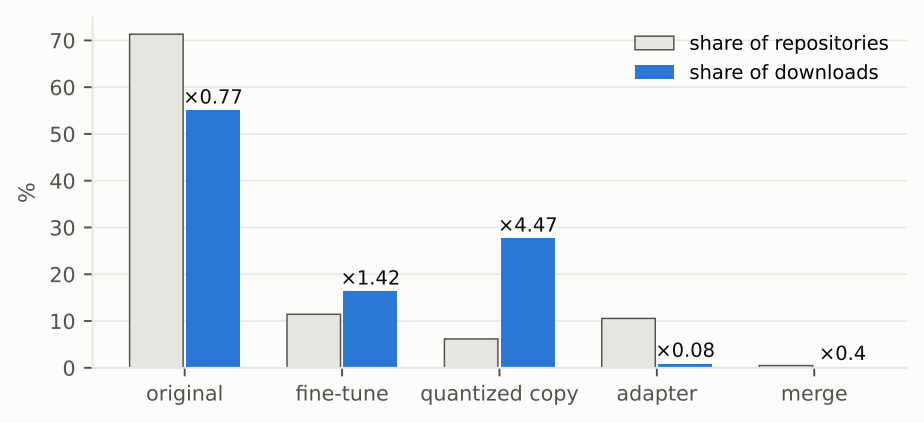



**Figure 9:** The whole Hub (3.2 M repositories, from the metadata dump): share of repositories and share of monthly downloads by the Hub's relation to a parent model; labels give each group's share of downloads divided by its share of repositories.

went from 2 to 8 and the share of parameters active per token fell from 28% to 11% (95% interval 10–13%; $\rho = -0.34$, $p < 10^{-7}$). Memory requirements depend on the total number of parameters and inference speed on the number of active ones; between 2024 and 2026, the ratio of total to active parameters grew from about 3.6 to about 9.

**Shape scaling of released models.** Across 965 dense original language models from 309 providers, width grows as $d \propto N^{0.40}$ (95% interval 0.38–0.42, $R^2 = 0.87$) and depth as $L \propto N^{0.28}$ (0.24–0.31, $R^2 = 0.56$), and the aspect ratio $d/L$, fitted directly, grows as $N^{0.13}$: larger released models are wider rather than proportionally deeper. For a dense transformer $N \approx 12d^2L$, so the exponents should satisfy $2 \cdot 0.40 + 0.28 \approx 1$; the sum is 1.08, slightly above one, which is consistent with embedding tables and wide feed-forward layers accounting for part of $N$. The vocabulary grows only weakly with size ($N^{0.14}$, 0.09–0.18, $R^2 = 0.07$) and is explained mostly by the tokenizer a model inherits. Tied input and output embeddings are common in small models (51% (43–59) below 1B parameters) and almost absent above 100B (2% (0–6)), with an odds ratio of 0.26 (0.21–0.32) per tenfold increase in size. One plausible explanation, which the census cannot test, is that the embedding table is a large share of a small model's parameters, so tying it saves the most memory there.

### 5.5.3 RQ9: Derivatives and lineage

**Where downloads go.** The Hub's base-model relation splits its 3.2 M repositories into originals (71%), fine-tunes (11%), adapters (11%), quantized copies (6%) and merges. Downloads are distributed differently (Figure 9): quantized copies receive 28% of all downloads, 4.47 times their share of repositories; originals receive 55%, fine-tunes 16%, and adapters almost none. Originals still receive the largest share, but close to half of all downloads go to a derivative. The largest publishers of the 899 popular quantized copies are `lmstudio-community` (135), `unsloth` (83), `Qwen` (53), `RedHatAI` (51), and 25% of these copies target Apple's MLX format.

**Architectural consistency of derivatives.** For every popular derivative whose single declared parent was also read, and where both anatomies were recovered, we compare the two. 73% (70–76) of fine-tunes ($n = 844$) and 84% (81–86) of quantized copies ($n = 722$) have exactly the parent's width, depth, head count, FFN width and vocabulary. Fine-tunes that differ change the vocabulary size (21% (18–23) of all fine-tunes) or the number of layers (13% (11–16)), and some change both. For quantized copies, the differences we

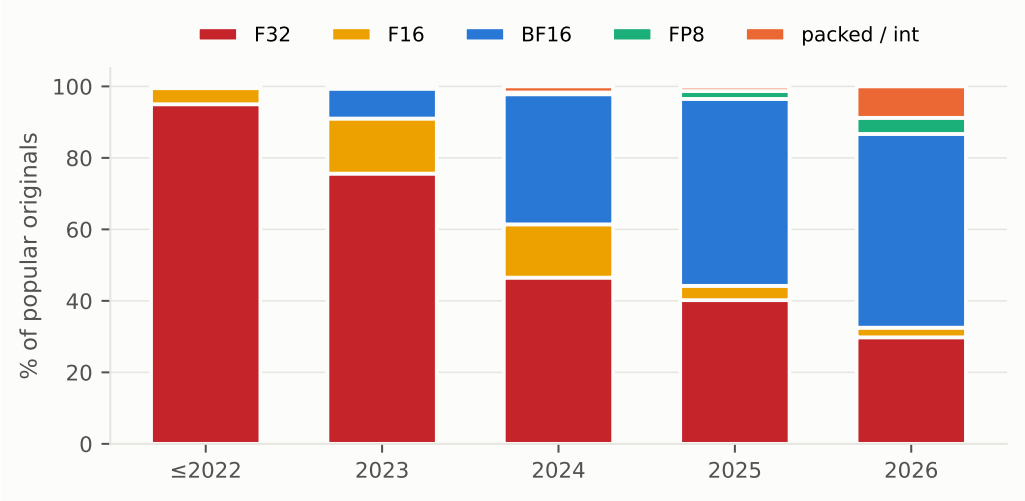



**Figure 10:** Dominant storage type of popular original releases by creation year (share of repositories).

inspected are mostly not changes to the model: smaller distilled models tagged as quantized copies of a larger one, GGUF repositories whose only safetensors file is an auxiliary component such as a VAE or text encoder, and padded feed-forward widths in AWQ checkpoints. In about three quarters of cases, therefore, a fine-tune keeps the parent's anatomy and changes only the weight values, so a comparison of headers can be used to verify a declared parent.

**Lineage.** Of the 4,826 parent links among popular repositories, 23% point to models by Qwen and 6% to the next provider (google). The Herfindahl index over parent owners, 0.065, is below common concentration thresholds because the remaining links spread over many owners, but Qwen alone accounts for more links than the next five owners combined.

### 5.5.4 RQ10: Metadata and what runs where

**The Hub's own metadata.** The Hub shows a parameter count computed from the number of stored elements. For formats that pack two values into a byte this is not the number of parameters: for 89% of 70 NVFP4 repositories the count differs from the logical one, by a median factor of 1.76. Over all readable popular repositories the count differs for 12.2% (11.3–13.1) and is missing for 24.4% (23.4–25.4); diffusion pipelines are counted by their denoiser alone. 11.9% (11.3–12.6) of popular repositories declare no license. Configurations disagree with the weights in 2.8% (2.3–3.3) of the 4,601 repositories whose configuration could be checked, and 60% of these disagreements sit in a sub-configuration (`audio_config`, `vision_config`) of a multimodal model. As in RQ4, a disagreement does not mean the configuration is incorrect; it means that a tool interpreting the field literally would compute incorrect shapes. Popular repositories ship a configuration more often than the Hub as a whole (15% without one, against 27.8% (25.3–30.3) of readable Hub repositories), but within the popular set, downloads do not differ measurably between repositories with and without one (Mann–Whitney, common-language effect size 0.52, $p = 0.093$).

**How weights are stored.** FP32 was the predominant storage type up to 2022 (95% (93–96) of popular original releases) and is now a minority (30% (27–33), odds ratio 0.45 (0.43–0.48) per year), replaced by BF16 (54% in 2026), FP8 and packed 4-bit formats (Figure 10). 13% (12–15) of popular models above 1B parameters still store weights in FP32, amounting to 9.8 TB of tensor data that BF16 storage would reduce by half.

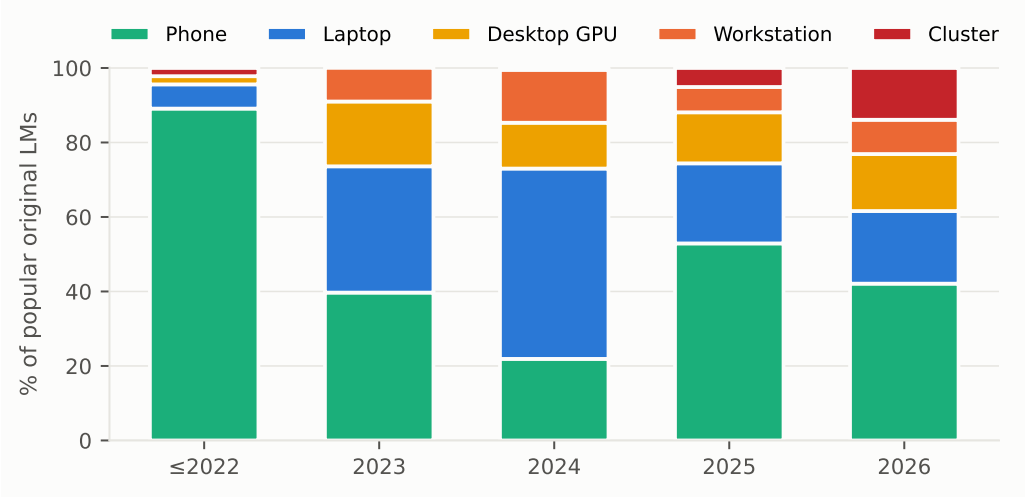



**Figure 11:** Smallest device class each popular original language model fits at 4-bit weights with an 8k-token 16-bit KV cache and 10% runtime overhead: phone (≤4.8 GB usable), 16 GB laptop (≤12 GB usable), desktop GPU (≤24 GB), workstation (≤80 GB), cluster.

**What runs where.** Because the census records the parameter count and the KV cache per token of every model, each model can be assigned to a device class. At 4-bit with an 8k context, 96% (85–99) of original language models created up to 2022 fit a 16 GB laptop, against 62% (57–66) of 2026 releases (odds ratio 0.76 (0.68–0.85) per year, $p < 10^{-5}$; Figure 11), and 14% of 2026 releases need more than 80 GB. Weighted by downloads, 45% of language-model use goes to models that fit a phone, somewhat more than their 40% share of releases.

**Open data and tools.** Both the labeled corpus and the census are released as datasets: the corpus with every manifest, its per-tensor labels, block lists and configuration; the census with one row per repository, the raw manifest and configuration of every readable one, the per-repository role scores and the sampling weights. Because the rules are code and the manifests are kept, the census can be re-scored offline by any later version of the method and re-run on a later snapshot to follow the ecosystem over time. A census page[4] presents these findings with a searchable table of models, a lookup that places any model among the popular ones and against common devices, and a memory map of parameters against KV cache per token. A *Deploy-Ready board* on the same page lists, for the leading providers' most used releases, the most downloaded language models and models built for Thai, and for any public repository submitted through a form, the smallest device class at 4-bit with an 8k context and five metadata checks: safetensors present, configuration present, configuration consistent with the weights, Hub parameter count correct, and license declared. The board reports deployability and metadata quality only; it does not rank models by output quality and is not a representative sample.

## 6 Discussion and limitations

**Labels and corpus.** Class names are an independent source of labels but not a perfect one. Some implementations place a position table inside a class called `...Embeddings` and others inside the encoder; some call a pooler a `...Head`; and linear-attention modules are named `...Attention` in some families and `...GatedDeltaNet` in others. We merged roles where the library is inconsistent and kept the fallback

[4] https://joessattes.github.io/ai-model-explorer/census/

list short, but per-role numbers for heads and adapters should be read with this in mind. We expect the headline numbers to be largely unaffected, because these roles account for a small share of the parameters.

All corpus manifests come from `transformers` implementations, which share naming habits. The shapes-only condition is the main protection against overfitting to those habits, and its 1.8-point drop suggests that the structural rules transfer. Default configurations are not trained checkpoints, but their shapes, which are all the method reads, follow the same rules. The checkpoint benchmark of Section 5.3 partly addresses this concern, since it uses real files published by many organizations. Fusion accuracy is measured against the same configuration that supplies the fields, so it validates the matching step and not the configuration; the consistency check is the independent part, and a richer check could model latent attention explicitly.

**Threats to validity.** Two further threats concern the corpus results. (1) The error analysis of Section 5.2 guided changes to the rules that were then checked on the same corpus, so corpus accuracy is not a held-out estimate; the same applies to the checkpoint benchmark and the census, since problems found in both also led to rule changes. Census repositories created after the rules were frozen will provide a prospective test. (2) The labels fall back to module names for loose tensors and cross-attention, so they are not entirely independent of the names the method reads.

Several threats apply to the census. Popularity is measured by one month of downloads, which may favor models that are downloaded by automated systems. The derivative relation comes from tags written by uploaders, which can be missing or incorrect; we use it as the Hub reports it, and the comparison of derivatives with their parents (RQ9) shows how often it is consistent with the weights. Family labels come from the method itself, with the accuracy of Table 8, and the pipeline filter that defines language models relies on a Hub tag. Year bins follow repository creation, not model release; the first bin absorbs every repository migrated in March 2022 and contains only models still popular today, so it mixes design change with survival. Intervals on the popular strata describe the release process under that view, not sampling error. The census reflects one snapshot and one sampling design; popular strata are read in full, but proportions for the Hub as a whole carry the sampling error stated with them, and repositories that ship only GGUF or adapter files are outside its scope. The Deploy-Ready board is a selection from the census plus whatever is submitted; it is provided as a service and is not used as evidence in this paper. A possible extension is a second track that scores other architecture-recovery tools on the corpus and census, as a shared benchmark.

**Implications.** Two practical consequences follow from the census. (1) A header-based check on the Hub would correct the parameter count of packed formats, flag configurations whose fields contradict the weights and catch derivatives whose declared parent has a different anatomy, at a cost of tens of kilobytes per repository. (2) The growing gap between total and active parameters, together with the falling KV cache per token, makes estimates of memory and speed based on parameter count alone increasingly unreliable.

**Scope.** GGUF and ONNX files are not yet supported, although both carry comparable metadata and GGUF alone accounts for 87% of the popular repositories the census could not read, and old non-zip PyTorch pickles are rejected. The order of operations within a block and the activation functions cannot be recovered, and compute estimates assume standard computation per role. The method reads only what a hub already serves publicly. It never downloads or executes weights, and the PyTorch reader interprets a fixed subset of pickle opcodes as data without resolving Python callables.

**Data and ethics.** The census reads only public metadata that the Hub serves to any client: file lists, configuration files and the first bytes of each weight file. Gated repositories were not accessed and are reported only as gated. The released dataset contains manifests and configurations, not weights, and names

publishers only as the public account names under which they publish. Requests stayed within the Hub's published rate limits.

# 7 Conclusion

A tensor manifest is small and almost always available, and it carries most of a model's architecture. With the module tree, a few dozen shape rules and width matching, we recover blocks, block types, regions, data paths and parameter counts for 510 architectures at 96.8% role accuracy, keep most of that accuracy without any names, and stay exact on parameter counts under current quantized formats. The same analysis also identifies its limits, since head geometry, masking and routing cannot be determined from the shapes. Taking them from the configuration closes most of the gap, and checking the configuration against the weights exposes the families whose fields mean something else. Applied to 12,974 Hub repositories, the same code measures the ecosystem. Mixtures of experts, grouped-query attention and multimodal input were adopted rapidly; the KV cache per token per billion parameters fell by more than an order of magnitude; larger models are wider rather than deeper; close to half of all downloads go to derivatives, which mostly keep the original anatomy; and the parameter count displayed by the Hub is incorrect for most packed 4-bit releases.

**Availability.** The explorer runs at `https://joessattes.github.io/ai-model-explorer/`, the census findings at `https://joessattes.github.io/ai-model-explorer/census/`, and the checkpoint benchmark at `https://joessattes.github.io/ai-model-explorer/bench.html`. The census dataset, with every raw manifest, is at `https://huggingface.co/datasets/JoesSattes/hub-weight-census`, the labeled corpus at `https://huggingface.co/datasets/JoesSattes/manifest-anatomy-corpus`, and the Deploy-Ready board at `https://joessattes.github.io/ai-model-explorer/census/#board`. Source code, the corpus builder and the evaluation scripts are at `https://github.com/JoesSattes/ai-model-explorer` under the MIT license. The corpus results are reproduced by `python research/build_corpus.py corpus.jsonl` followed by `node research/eval.js corpus.jsonl` with the flags `--obf --nolex`, `--quant nvfp4`, `--v1` and `--cfg`; the checkpoint results by `node research/realbench.js export.jsonl corpus.jsonl`. The corpus used here is `research/data/corpus-510.jsonl.gz`; the census is reproduced by `frame.py`, `sample.py`, `crawl.py`, `summarize.py`, `insights.py` and `roles.js` in `research/hub-census`, or re-scored from released manifests with `rescore.py`.

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# A Corpus construction

The model class for each configuration type is chosen in this order: image-text-to-text, multimodal LM, speech sequence-to-sequence, sequence-to-sequence LM, TDT, RNN-T, causal LM, CTC, masked LM, image classification, text-to-waveform, object detection, base model. Instantiation uses `_from_config` under `torch.device("meta")` with a 90-second limit. Of the 240 types that do not instantiate, 177 are sub-configurations without a model class of their own, such as vision or audio towers registered separately,

and the rest fail on required sub-configurations or optional dependencies. Manifests are deduplicated by the exact list of names and shapes, and tied parameters are kept once under the name PyTorch reports first.

Each record stores the manifest (name, dtype and shape of every tensor), the per-tensor ground-truth role, the module lists with their item classes, the parameter total, and the serialized configuration. Table 10 gives the families assigned by our method; these assignments approximate the composition of the corpus. "Encoder" covers bidirectional encoders, vision backbones and any decoder with tied embeddings whose surroundings look like an encoder's (Section 3.2 gives the cues; Table 8 measures how often they hold on published checkpoints). A "prefix LM" is a decoder that reads the output of another component, usually an image or audio encoder, as a prefix to its text.

| | encoder | decoder | prefix LM | enc–dec | transducer | diffusion | unknown |
|---|---|---|---|---|---|---|---|
| Models | 356 | 93 | 46 | 8 | 3 | 2 | 2 |

**Table 10:** Families assigned to the 510 corpus models. A prefix LM is a decoder that takes the output of an encoder, usually for images or audio, as a prefix.

# B Ground-truth labeling

A tensor's label comes from the classes of the modules on its path. If any ancestor class matches the MoE pattern, the label is mixture of experts. Otherwise the nearest ancestor whose lower-cased class name matches one of the patterns in Table 11, tried in the order shown, gives the label; an attention match whose module path contains `encoder_attn` or `cross_attn` becomes cross-attention. Tensors that no class matches fall back to their module name: `lm_head`, `classifier`, `score` or `joint` give head; `fc`*n*, `mlp`, `dense`, `w`*n*, `c_fc` and `c_proj` give feed-forward; `conv`, `patch` or `subsampl` outside the layer lists give input stem; `merger`, `projector` or `connector` give adapter. Everything else is *other* and is excluded from scoring.

| Label | Class-name pattern (suffix of the lower-cased class name) |
|---|---|
| MoE (any ancestor) | `sparsemoeblock`, `moeblock`, `sparsemoe`, `moe`, `moelayer`, `experts`, `expert`, `router`, `topkrouter`, `moegate`, `sparsemlp`, `routedexperts` |
| head | `predictionheadtransform`, `lmpredictionhead`, `predictionhead`, `lmhead`, `pooler`, `classificationhead`, `mlmhead`, `predictions` |
| attention | `selfoutput`, `attentionoutput` |
| norm | `rmsnorm`, `layernorm`, `groupnorm`, `batchnorm1d/2d`, `norm` |
| cross-attention | `crossattention`, `encdecattention`, `crossattn` |
| SSM / linear attn. | `mamba(2)mixer`, `mixer`, `mamba`, `ssm`, `deltanet`, `gateddeltanet`, `linearattention`, `lightningattention`, `rwkv*`, `recurrentblock`, `rglru`, `xlstm*`, `retention`, `hgrn`, `gla` |
| attention | `attention`, `attn`, `selfattention`, `multiheadattention`, `sdpaattention` |
| recurrent cell | `lstm`, `gru`, `rnn`, `lstmcell` |
| adaLN modulation | `adalayernorm*`, `adaln*`, `modulation`, `timestepembedding`, `timestepembedder` |
| convolution module | `convolutionmodule`, `convmodule`, `conformerconvolution*`, `convblock`, `depthwiseconv*` |
| feed-forward | `mlp`, `feedforward`, `ffn`, `intermediate`, `output`, `swiglu`, `geglu`, `densereludense`, `densegatedactdense` |
| input stem | `embedding(s)`, `embed`, `patchembed*` |

**Table 11:** Class-name patterns used for the ground truth, in the order they are tried.

# C Lexicon and thresholds

The optional lexicon consists of eight name patterns, one per region role, plus patterns for the query, key, value, output, gate, up and down projections inside a region. It is consulted only in the following four cases: to choose between two region labels whose shape signatures are equally strong; to tell a token table from an untied output head when both have shape $[V, d]$ and neither position nor nesting decides; to accept a recurrent cell, which additionally requires `ih`/`hh` or a cell name; and to name loose projectors whose shapes match two components. In the shapes-only condition all four fall back to shape rules or to the label *other*.

The numeric thresholds are the following. Depthwise convolutions count as SSM convolutions up to kernel size 8 and as conformer convolutions above it or when pointwise convolutions are present. Per-head gates have at most 64 rows. A fused feed-forward requires $f > 1.3d$. A router may have up to eight more rows than there are experts. Cross-attention requires two projections from a foreign width of at least 128. adaLN requires a conditioning width of at least 32. Adapters allow pixel-shuffle factors $k \in \{2, 3, 4\}$. A transducer joint may have up to 16 outputs more than the label table. Without a configuration, the head size is 128 for $d \geq 2048$ and 64 otherwise. A multi-token-prediction layer is the last item of a stack that holds a $[d, 2d]$ matrix and at least two more width-$d$ vectors than the item before it.

# D Synthetic quantized storage

The rewrite applies to every 2-D tensor named `*.weight` whose input dimension is a multiple of 32 and output dimension a multiple of 8, excluding embeddings, heads and norms. NVFP4 replaces $[o, i]$ by `U8` $[o, i/2]$, `F8_E4M3` $[o, i/16]$ and two `F32` scalars. FP8 uses `F8_E4M3` $[o, i]$ and `F32` $[o, 1]$. GPTQ uses `qweight` as `I32` $[i/8, o]$, `qzeros` $[i/128, o/8]$, `scales` as `F16` $[i/128, o]$ and `g_idx` $[i]$. Under name hashing these suffixes are hashed too, so recognition relies on dtypes and shape ratios. On real checkpoints the loader also handles MXFP4 (`_blocks` and `_scales` pairs), MLX (`U32` words with `scales` and `biases`), and bitsandbytes NF4, whose state tensors are stored under `<module>.weight.*`.

# E Corpus results per model

Table 12 lists the most frequent confusions and Table 13 the role accuracy of every corpus model, with and without names.

| Ground truth → predicted | Tensors | Models (examples) |
|---|---|---|
| attn → conv | 1387 | cvt, efficientloftr |
| ssm → attn | 1130 | glm5_next |
| ffn → other | 871 | albert, cvt |
| attn → ssm | 676 | edgetam_video |
| ffn → head | 641 | blt, canine, cohere2_vision |
| attn → norm | 593 | mistral_large4 |
| attn → other | 553 | albert |
| attn → head | 328 | blt |
| ffn → conv | 302 | d_fine, deimv2 |
| ffn → attn | 284 | evolla |
| attn → ffn | 198 | bridgetower |
| norm → head | 158 | aimv2, aimv2_vision_model |
| embed → attn | 140 | gemma3n_audio |
| embed → conv | 129 | cvt |
| ssm → conv | 120 | minimax |
| cross → attn | 116 | blt |

**Table 12:** The most frequent confusions of the full method on the corpus, counted in tensors.

**Table 13:** Per-model role accuracy (%) on the corpus: full method / shapes only. Model types are the `transformers` configuration names.

| Type | Full | Shapes | Type | Full | Shapes |
|---|---|---|---|---|---|
| afmoe | 100.0 | 100.0 | minimax | 97.1 | 97.1 |
| aimv2 | 100.0 | 100.0 | minimax_m2 | 100.0 | 100.0 |
| aimv2_vision_model | 100.0 | 100.0 | minimax_m3_vl | 100.0 | 100.0 |
| albert | 0.0 | 0.0 | minimax_m3_vl_text | 100.0 | 100.0 |
| align | 100.0 | 100.0 | ministral3 | 100.0 | 100.0 |
| altclip | 100.0 | 100.0 | mistral | 100.0 | 100.0 |
| apertus | 100.0 | 100.0 | mistral3 | 100.0 | 100.0 |
| arcee | 100.0 | 100.0 | mistral4 | 100.0 | 100.0 |
| aria_text | 100.0 | 100.0 | mistral_large4 | 100.0 | 100.0 |
| audio-spectrogram-transformer | 100.0 | 100.0 | mistral_large4_text | 100.0 | 100.0 |
| audioflamingo3 | 100.0 | 100.0 | mixtral | 100.0 | 100.0 |
| audioflamingo3_encoder | 100.0 | 100.0 | mlcd_vision_model | 100.0 | 100.0 |
| axk1 | 100.0 | 100.0 | mllama | 100.0 | 91.5 |
| axk2 | 100.0 | 100.0 | mm-grounding-dino | 88.5 | 86.0 |
| bailing_hybrid | 100.0 | 100.0 | mobilebert | 38.1 | 36.9 |
| bamba | 100.0 | 100.0 | mobilenet_v1 | 100.0 | 99.9 |
| bark | 81.9 | 94.0 | mobilenet_v2 | 100.0 | 99.9 |
| bart | 100.0 | 100.0 | mobilevit | 100.0 | 100.0 |
| beam | 100.0 | 100.0 | mobilevitv2 | 39.8 | 39.8 |
| beit | 100.0 | 100.0 | modernbert | 100.0 | 100.0 |
| bert | 100.0 | 100.0 | modernbert-decoder | 100.0 | 100.0 |
| bert-generation | 100.0 | 100.0 | modernvbert | 97.6 | 97.6 |
| big_bird | 100.0 | 100.0 | molmo2 | 98.6 | 91.5 |
| bigbird_pegasus | 98.5 | 98.5 | molmo2_text | 100.0 | 100.0 |
| biogpt | 100.0 | 100.0 | moonshine | 100.0 | 100.0 |
| bit | 100.0 | 100.0 | moshi | 100.0 | 98.1 |
| bitnet | 100.0 | 100.0 | moss_transcribe_diarize | 100.0 | 99.9 |
| blenderbot | 100.0 | 100.0 | moss_transcribe_diarize_audio | 100.0 | 100.0 |
| blenderbot-small | 99.4 | 99.4 | mpnet | 100.0 | 100.0 |
| blip | 100.0 | 100.0 | mpt | 100.0 | 100.0 |
| blip-2 | 100.0 | 100.0 | mra | 100.0 | 100.0 |
| blip_2_qformer | 100.0 | 100.0 | mt5 | 100.0 | 96.3 |
| bloom | 100.0 | 100.0 | muse_glimmer | 99.9 | 99.9 |
| blt | 98.7 | 98.7 | muse_glimmer_assistant | 100.0 | 100.0 |
| bridgetower | 82.6 | 82.6 | muse_glimmer_text | 100.0 | 100.0 |

| Type | Full | Shapes | Type | Full | Shapes |
|---|---|---|---|---|---|
| bros | 100.0 | 100.0 | muse_glimmer_vision | 99.9 | 99.9 |
| camembert | 100.0 | 78.6 | muse_spark | 100.0 | 100.0 |
| canary | 100.0 | 100.0 | muse_spark_audio | 91.7 | 88.3 |
| canine | 89.3 | 79.7 | muse_spark_text | 100.0 | 100.0 |
| chinese_clip | 100.0 | 100.0 | mvp | 100.0 | 100.0 |
| chinese_clip_vision_model | 100.0 | 100.0 | nanochat | 100.0 | 100.0 |
| clap | 100.0 | 100.0 | nemo_speechlm | 100.0 | 100.0 |
| clip | 100.0 | 100.0 | nemotron3_5_asr | 99.8 | 96.8 |
| clip_text_model | 100.0 | 99.9 | nemotron3_5_transcribe_asr_audio | 100.0 | 100.0 |
| clipseg | 100.0 | 100.0 | nemotron3_5_transcribe_audio | 100.0 | 100.0 |
| clvp | 100.0 | 99.9 | nemotron3_5_transcribe_word_timestamp | 100.0 | 100.0 |
| codegen | 100.0 | 96.6 | nemotron3_diarization | 100.0 | 100.0 |
| cohere | 100.0 | 100.0 | nemotron3_diarization_audio | 100.0 | 100.0 |
| cohere2_moe | 100.0 | 100.0 | nemotron_asr_streaming | 99.9 | 99.6 |
| cohere2_vision | 100.0 | 99.5 | nemotron_asr_streaming_encoder | 100.0 | 100.0 |
| cohere_asr | 99.9 | 99.9 | nemotron_h | 100.0 | 100.0 |
| cohere_compass_vision | 99.5 | 72.9 | nemotron_h_omni | 100.0 | 100.0 |
| convbert | 100.0 | 100.0 | neomme | 28.8 | 0.9 |
| convnext | 99.6 | 99.5 | neucodec | 100.0 | 100.0 |
| convnextv2 | 100.0 | 99.9 | nllb-moe | 100.0 | 6.1 |
| cosmos3_edge | 100.0 | 100.0 | nomic_bert | 100.0 | 100.0 |
| cosmos3_edge_text | 100.0 | 100.0 | nystromformer | 100.0 | 100.0 |
| cosmos3_edge_vision | 99.8 | 99.8 | olmo | 100.0 | 100.0 |
| cosmos3_omni | 100.0 | 98.8 | olmo2 | 100.0 | 100.0 |
| cpmant | 100.0 | 98.7 | olmo_hybrid | 100.0 | 100.0 |
| csm | 99.7 | 99.7 | olmoe | 100.0 | 100.0 |
| ctrl | 100.0 | 100.0 | oneformer | 99.9 | 85.5 |
| cvt | 95.7 | 95.7 | openai-gpt | 100.0 | 100.0 |
| cwm | 100.0 | 100.0 | openai_privacy_filter | 100.0 | 100.0 |
| d_fine | 92.0 | 92.0 | opt | 100.0 | 100.0 |
| dac | 99.9 | 99.9 | ovis2 | 100.0 | 100.0 |
| data2vec-audio | 96.0 | 96.0 | owlv2 | 100.0 | 100.0 |
| data2vec-text | 100.0 | 100.0 | owlvit | 100.0 | 100.0 |
| data2vec-vision | 100.0 | 100.0 | paddleocr_vl | 100.0 | 100.0 |
| deberta | 100.0 | 100.0 | paligemma | 100.0 | 100.0 |
| deberta-v2 | 100.0 | 100.0 | parakeet_ctc | 100.0 | 100.0 |
| decision_transformer | 89.5 | 89.5 | parakeet_encoder | 100.0 | 100.0 |
| deepseek_v2 | 100.0 | 100.0 | parakeet_rnnt | 99.8 | 98.2 |
| deepseek_v3 | 100.0 | 100.0 | parakeet_tdt | 99.8 | 98.2 |
| deepseek_v32 | 100.0 | 100.0 | patchtsmixer | 10.3 | 10.3 |
| deepseek_v4 | 100.0 | 100.0 | patchtst | 100.0 | 100.0 |
| deepseek_vl | 99.9 | 99.9 | pe_audio | 100.0 | 100.0 |
| deepseek_vl_hybrid | 99.9 | 99.9 | pe_audio_encoder | 100.0 | 100.0 |
| deimv2 | 91.4 | 91.4 | pegasus | 99.5 | 99.5 |
| deit | 100.0 | 100.0 | pegasus_x | 100.0 | 100.0 |
| depth_pro | 100.0 | 100.0 | perceiver | 96.4 | 95.7 |
| dia | 100.0 | 100.0 | persimmon | 100.0 | 100.0 |
| diffllama | 100.0 | 100.0 | phi | 100.0 | 100.0 |
| dinov2 | 100.0 | 100.0 | phi3 | 100.0 | 100.0 |
| dinov2_with_registers | 99.8 | 99.8 | phi4_multimodal | 99.0 | 99.0 |
| dinov3_convnext | 100.0 | 100.0 | phimoe | 100.0 | 100.0 |
| dinov3_vit | 100.0 | 100.0 | pi0 | 84.9 | 99.9 |
| distilbert | 100.0 | 100.0 | pix2struct | 98.7 | 61.3 |
| doge | 100.0 | 100.0 | pixio | 100.0 | 100.0 |
| donut-swin | 100.0 | 100.0 | pixtral | 100.0 | 100.0 |
| dpr | 100.0 | 100.0 | plbart | 100.0 | 100.0 |
| dpt | 100.0 | 100.0 | pocket_tts | 95.3 | 95.3 |
| edgetam_video | 90.8 | 89.3 | pocket_tts_audio_tokenizer | 100.0 | 100.0 |
| efficientloftr | 78.6 | 78.6 | pocket_tts_audio_tokenizer_decoder | 100.0 | 100.0 |
| efficientnet | 100.0 | 100.0 | pocket_tts_audio_tokenizer_encoder | 100.0 | 100.0 |
| electra | 99.3 | 70.3 | poolformer | 0.1 | 0.1 |
| encodec | 100.0 | 100.0 | pop2piano | 100.0 | 89.3 |
| encoder-decoder | 90.5 | 81.0 | pp_doclayout_v2 | 96.6 | 96.6 |
| ernie | 100.0 | 100.0 | pp_doclayout_v3 | 96.5 | 96.5 |
| ernie4_5 | 100.0 | 100.0 | pp_doclayout_v4 | 96.8 | 96.8 |
| ernie4_5_moe | 100.0 | 100.0 | pp_formulanet | 100.0 | 99.1 |

| Type | Full | Shapes | Type | Full | Shapes |
|---|---|---|---|---|---|
| ernie4_5_vl_moe | 99.5 | 99.5 | pp_lcnet | 100.0 | 99.6 |
| esmc | 100.0 | 100.0 | pp_ocrv5_mobile_det | 100.0 | 100.0 |
| eurobert | 100.0 | 100.0 | pp_ocrv5_mobile_rec | 100.0 | 100.0 |
| evolla | 84.5 | 89.7 | pp_ocrv5_server_rec | 100.0 | 100.0 |
| exaone4 | 100.0 | 100.0 | pp_ocrv6_small_det | 100.0 | 100.0 |
| exaone4_5 | 99.6 | 97.9 | pp_ocrv6_small_rec | 100.0 | 100.0 |
| exaone4_5_vision | 98.1 | 90.4 | prophetnet | 100.0 | 100.0 |
| exaone_moe | 100.0 | 100.0 | pvt | 82.5 | 82.5 |
| falcon | 100.0 | 100.0 | pvt_v2 | 53.0 | 53.0 |
| falcon_h1 | 100.0 | 100.0 | qianfan_ocr | 100.0 | 99.7 |
| falcon_mamba | 100.0 | 75.8 | qianfan_ocr_vision | 100.0 | 100.0 |
| fastspeech2_conformer | 99.7 | 99.7 | qwen2 | 100.0 | 100.0 |
| fastspeech2_conformer_with_hifigan | 99.9 | 99.9 | qwen2_5_omni | 99.7 | 98.4 |
| flaubert | 100.0 | 100.0 | qwen2_5_omni_thinker | 99.5 | 97.7 |
| flava | 100.0 | 99.5 | qwen2_5_vl | 99.9 | 99.7 |
| flex_olmo | 100.0 | 100.0 | qwen2_5_vl_text | 100.0 | 100.0 |
| flint | 100.0 | 100.0 | qwen2_5_vl_vision | 98.3 | 91.7 |
| florence2 | 95.1 | 93.0 | qwen2_audio | 100.0 | 100.0 |
| fnet | 99.3 | 99.3 | qwen2_moe | 100.0 | 100.0 |
| focalnet | 86.0 | 62.6 | qwen2_vl | 100.0 | 99.9 |
| fsmt | 90.2 | 90.2 | qwen3 | 100.0 | 100.0 |
| fun_asr_nano | 73.7 | 73.7 | qwen3_5 | 100.0 | 99.6 |
| fun_asr_nano_encoder | 2.9 | 2.9 | qwen3_5_moe | 100.0 | 99.9 |
| funnel | 100.0 | 100.0 | qwen3_5_moe_text | 100.0 | 100.0 |
| fuyu | 100.0 | 100.0 | qwen3_5_moe_vision | 99.4 | 91.1 |
| gemma | 100.0 | 100.0 | qwen3_5_text | 100.0 | 100.0 |
| gemma2 | 100.0 | 100.0 | qwen3_asr | 99.8 | 99.8 |
| gemma3 | 99.7 | 99.7 | qwen3_asr_encoder | 100.0 | 100.0 |
| gemma3_text | 100.0 | 100.0 | qwen3_moe | 100.0 | 100.0 |
| gemma3n_audio | 83.4 | 83.4 | qwen3_next | 100.0 | 100.0 |
| gemma3n_text | 98.1 | 63.3 | qwen3_omni_moe_thinker | 99.9 | 99.1 |
| gemma4_audio | 99.0 | 99.0 | qwen3_vl_moe | 100.0 | 98.9 |
| gemma4_vision | 100.0 | 100.0 | qwen3_vl_moe_text | 100.0 | 100.0 |
| git | 100.0 | 100.0 | qwen3_vl_text | 100.0 | 100.0 |
| glm | 100.0 | 100.0 | radio | 99.8 | 99.8 |
| glm4 | 100.0 | 100.0 | recurrent_gemma | 81.2 | 81.2 |
| glm46v | 100.0 | 100.0 | regnet | 100.0 | 100.0 |
| glm4_moe | 100.0 | 100.0 | rembert | 57.8 | 51.4 |
| glm4_moe_lite | 100.0 | 100.0 | resnet | 100.0 | 100.0 |
| glm4v_moe | 100.0 | 100.0 | rf_detr | 99.6 | 99.5 |
| glm4v_moe_text | 100.0 | 100.0 | roberta | 100.0 | 100.0 |
| glm4v_moe_vision | 99.9 | 79.2 | roberta-prelayernorm | 100.0 | 100.0 |
| glm4v_text | 100.0 | 100.0 | roc_bert | 100.0 | 89.8 |
| glm5_next | 98.5 | 98.5 | roformer | 99.9 | 99.9 |
| glm5_next_text | 98.5 | 98.5 | rt_detr | 94.9 | 94.9 |
| glm5_next_vision | 100.0 | 89.2 | rt_detr_v2 | 94.9 | 94.9 |
| glm_image | 100.0 | 99.7 | rwkv | 100.0 | 100.0 |
| glm_image_text | 100.0 | 100.0 | sam | 97.4 | 95.8 |
| glm_image_vision | 100.0 | 100.0 | sam2 | 90.7 | 89.0 |
| glm_image_vqmodel | 100.0 | 15.8 | sam2_hiera_det_model | 92.7 | 92.7 |
| glm_moe_dsa | 100.0 | 100.0 | sam2_video | 80.4 | 79.0 |
| glm_ocr | 100.0 | 97.1 | sam2_vision_model | 92.8 | 92.8 |
| glm_ocr_text | 100.0 | 100.0 | sam3 | 99.8 | 99.4 |
| glm_ocr_vision | 100.0 | 94.6 | sam3_lite_text | 98.9 | 98.3 |
| glmasr | 100.0 | 100.0 | sam3_lite_text_text_model | 93.1 | 93.1 |
| glmasr_encoder | 100.0 | 100.0 | sam3_tracker | 99.8 | 99.7 |
| glpn | 75.0 | 75.0 | sam3_tracker_video | 98.8 | 98.7 |
| got_ocr2 | 100.0 | 99.8 | sam3_video | 99.1 | 98.7 |
| gpt-sw3 | 100.0 | 100.0 | sam3_vision_model | 100.0 | 100.0 |
| gpt_bigcode | 100.0 | 100.0 | sam3_vit_model | 100.0 | 100.0 |
| gpt_neo | 99.7 | 99.7 | sam_hq | 97.1 | 95.5 |
| gpt_neox | 100.0 | 98.5 | sam_hq_vision_model | 100.0 | 100.0 |
| gpt_neox_japanese | 100.0 | 100.0 | sapiens2 | 100.0 | 100.0 |
| gpt_oss | 100.0 | 100.0 | seamless_m4t | 96.4 | 96.4 |
| gptj | 100.0 | 96.6 | seamless_m4t_v2 | 96.4 | 96.4 |
| gradium_phonon_tts | 92.0 | 92.0 | seed_oss | 100.0 | 100.0 |

| Type | Full | Shapes | Type | Full | Shapes |
|---|---|---|---|---|---|
| granite | 100.0 | 100.0 | segformer | 75.0 | 75.0 |
| granite_speech | 100.0 | 100.0 | seggpt | 100.0 | 100.0 |
| granite_speech5_ctc | 99.5 | 76.5 | sew | 100.0 | 100.0 |
| granite_speech5_encoder | 99.5 | 76.5 | sew-d | 100.0 | 100.0 |
| granite_swa | 100.0 | 100.0 | siglip | 92.4 | 92.4 |
| granitemoe | 100.0 | 100.0 | siglip2 | 91.7 | 91.7 |
| granitemoe_swa | 100.0 | 100.0 | siglip2_vision_model | 91.7 | 91.7 |
| granitemoehybrid | 98.9 | 98.9 | siglip_vision_model | 92.4 | 92.4 |
| grounding-dino | 88.5 | 86.0 | smollm3 | 100.0 | 100.0 |
| groupvit | 85.9 | 85.9 | solar_open | 100.0 | 100.0 |
| gte | 100.0 | 100.0 | speech-encoder-decoder | 100.0 | 100.0 |
| helium | 100.0 | 100.0 | speech_to_text | 100.0 | 100.0 |
| hgnet_v2 | 100.0 | 100.0 | speecht5 | 100.0 | 99.9 |
| hiera | 96.1 | 96.1 | splinter | 100.0 | 100.0 |
| higgs_audio_v2 | 100.0 | 99.5 | squeezebert | 56.3 | 56.2 |
| hrm_text | 100.0 | 100.0 | stablelm | 100.0 | 100.0 |
| hubert | 100.0 | 100.0 | starcoder2 | 100.0 | 100.0 |
| hunyuan_vl_vision | 94.7 | 94.7 | step3p7 | 100.0 | 100.0 |
| hy_v3 | 100.0 | 100.0 | swiftformer | 71.7 | 71.7 |
| hy_v4 | 100.0 | 100.0 | swin | 100.0 | 100.0 |
| hyperclovax | 100.0 | 100.0 | swin2sr | 98.3 | 98.3 |
| hyperclovax_vision_v2 | 99.5 | 97.4 | swinv2 | 100.0 | 100.0 |
| ibert | 11.4 | 11.4 | switch_transformers | 100.0 | 45.7 |
| idefics | 100.0 | 100.0 | t5 | 100.0 | 89.6 |
| idefics2 | 99.0 | 99.0 | t5gemma | 89.5 | 89.5 |
| idefics3 | 99.7 | 99.7 | t5gemma2 | 99.9 | 99.9 |
| idefics3_vision | 100.0 | 100.0 | tapas | 100.0 | 100.0 |
| ijepa | 100.0 | 100.0 | textnet | 100.0 | 100.0 |
| imagegpt | 99.0 | 99.0 | timesfm | 100.0 | 100.0 |
| inkling_mm_model | 100.0 | 100.0 | timesfm2_5 | 100.0 | 100.0 |
| inkling_text | 100.0 | 100.0 | timesformer | 100.0 | 75.2 |
| instructblip | 100.0 | 100.0 | tipsv2 | 100.0 | 100.0 |
| internvl | 100.0 | 99.7 | tipsv2_text_model | 100.0 | 100.0 |
| internvl_vision | 100.0 | 100.0 | tipsv2_vision_model | 100.0 | 100.0 |
| jais2 | 100.0 | 100.0 | trocr | 100.0 | 100.0 |
| jamba | 100.0 | 100.0 | tvp | 99.9 | 99.9 |
| janus | 99.8 | 99.8 | udop | 99.7 | 86.2 |
| jetmoe | 97.7 | 97.7 | ultrabert | 100.0 | 100.0 |
| jina_embeddings_v3 | 100.0 | 54.3 | ultravbert | 99.9 | 99.9 |
| kimi_k25 | 100.0 | 100.0 | umt5 | 100.0 | 96.3 |
| kimi_k25_vision | 100.0 | 100.0 | unispeech | 100.0 | 100.0 |
| kimi_linear | 99.6 | 99.6 | unispeech-sat | 100.0 | 100.0 |
| kolibri1 | 100.0 | 100.0 | univnet | 100.0 | 100.0 |
| kosmos-2 | 98.9 | 98.9 | unlimited_ocr | 100.0 | 100.0 |
| kosmos-2.5 | 99.1 | 98.7 | uvdoc | 100.0 | 100.0 |
| kyutai_speech_to_text | 99.7 | 99.7 | vaultgemma | 100.0 | 100.0 |
| laguna | 100.0 | 100.0 | vibevoice | 98.4 | 98.2 |
| lasr_ctc | 100.0 | 100.0 | vibevoice_acoustic_tokenizer | 100.0 | 100.0 |
| lasr_encoder | 100.0 | 100.0 | vibevoice_acoustic_tokenizer_decoder | 100.0 | 100.0 |
| layoutlm | 100.0 | 100.0 | vibevoice_acoustic_tokenizer_encoder | 100.0 | 100.0 |
| layoutlmv3 | 100.0 | 99.6 | vibevoice_asr | 100.0 | 99.7 |
| led | 99.8 | 99.8 | video_llama_3 | 100.0 | 100.0 |
| levit | 100.0 | 100.0 | video_llama_3_vision | 100.0 | 100.0 |
| lfm2 | 100.0 | 100.0 | video_llava | 100.0 | 100.0 |
| lfm2_vl | 99.7 | 99.7 | videomae | 100.0 | 100.0 |
| lightglue | 64.0 | 64.0 | videoprism | 100.0 | 100.0 |
| lighton_ocr | 99.6 | 99.6 | videoprism_vision_model | 100.0 | 100.0 |
| lilt | 99.9 | 99.3 | vilt | 100.0 | 100.0 |
| llama4 | 100.0 | 100.0 | vipllava | 100.0 | 99.8 |
| llama4_text | 100.0 | 100.0 | vision-encoder-decoder | 100.0 | 100.0 |
| llava | 100.0 | 100.0 | vision-text-dual-encoder | 100.0 | 100.0 |
| llava_next | 100.0 | 100.0 | visual_bert | 99.6 | 99.6 |
| llava_onevision | 100.0 | 100.0 | vit | 100.0 | 100.0 |
| longcat_flash | 100.0 | 100.0 | vit_mae | 100.0 | 100.0 |
| longformer | 100.0 | 100.0 | vitdet | 100.0 | 100.0 |
| longt5 | 100.0 | 89.6 | vits | 72.8 | 72.8 |

| Type | Full | Shapes | Type | Full | Shapes |
|---|---|---|---|---|---|
| luke | 99.9 | 46.8 | vivit | 100.0 | 100.0 |
| lw_detr | 98.9 | 95.5 | vjepa2 | 99.9 | 99.9 |
| lxmert | 100.0 | 100.0 | voxtral | 100.0 | 100.0 |
| m2m_100 | 100.0 | 100.0 | voxtral_realtime | 99.9 | 99.9 |
| mamba2 | 98.2 | 98.2 | voxtral_realtime_encoder | 100.0 | 100.0 |
| marian | 99.5 | 99.5 | wav2vec2 | 100.0 | 100.0 |
| markuplm | 97.0 | 91.2 | wavlm | 100.0 | 100.0 |
| mask2former | 99.9 | 99.9 | whisper | 100.0 | 100.0 |
| maskformer | 100.0 | 100.0 | xclip | 98.3 | 81.7 |
| maskformer-swin | 100.0 | 100.0 | xcodec | 99.3 | 99.3 |
| mbart | 100.0 | 100.0 | xcodec2 | 100.0 | 100.0 |
| megatron-bert | 100.0 | 100.0 | xglm | 100.0 | 100.0 |
| mellum | 100.0 | 100.0 | xlm-roberta-xl | 100.0 | 100.0 |
| mgp-str | 100.0 | 100.0 | xlnet | 44.1 | 35.0 |
| mimi | 99.5 | 99.5 | xlstm | 37.3 | 34.3 |
| mimi_decoder | 100.0 | 100.0 | xmod | 69.6 | 69.6 |
| mimi_encoder | 100.0 | 100.0 | yolos | 100.0 | 100.0 |
| mimo_v2_flash | 100.0 | 100.0 | yoso | 100.0 | 100.0 |
| minicpm3 | 100.0 | 100.0 | youtu | 100.0 | 100.0 |
| minicpm_ocr | 99.6 | 99.1 | zamba | 100.0 | 100.0 |
| minicpm_ocr_vision | 98.3 | 70.4 | zamba2 | 78.8 | 78.8 |
| minicpmv4_6 | 99.6 | 99.2 | zaya | 97.5 | 97.5 |

# F Hub benchmark: all checkpoints

The benchmark page is https://joessattes.github.io/ai-model-explorer/bench.html; its model list is in bench/models.js in the repository, and the export analyzed here is research/data/hub-bench-191.jsonl.gz. Table 14 lists the repositories that could not be read and Table 15 the result for each of the 191 that could. Header sizes are those of the original run, before the loader was modified to skip duplicate files, so they are larger than necessary for the few repositories that ship several copies of the same weights.

| Repository | Group | Reason |
|---|---|---|
| CohereLabs/c4ai-command-r7b-12-2024 | Dense LM | gated: the license must be accepted on the Hub first |
| NousResearch/Meta-Llama-3.1-405B | Dense LM | gated: the license must be accepted on the Hub first |
| ai21labs/AI21-Jamba-Mini-1.7 | SSM / hybrid | gated: the license must be accepted on the Hub first |
| black-forest-labs/FLUX.1-schnell | Image / video gen. | gated: the license must be accepted on the Hub first |
| google/gemma-3-270m | Dense LM | gated: the license must be accepted on the Hub first |
| google/paligemma2-3b-pt-224 | Vision–language | gated: the license must be accepted on the Hub first |
| pyannote/segmentation-3.0 | Speech / audio | gated: the license must be accepted on the Hub first |
| Helsinki-NLP/opus-mt-en-de | Seq2seq | old non-zip PyTorch pickle (not read by design) |
| google/electra-base-discriminator | Encoder | old non-zip PyTorch pickle (not read by design) |
| facebook/mms-1b-all | Speech / audio | rate limited: one adapter file per language; the loader now skips adapters |

**Table 14:** The ten repositories that could not be read.

**Table 15:** All 191 checkpoints read from the Hub, by group. *Params* is our logical count and *Hub* the count on the model page; the next column marks agreement (=), a difference (≠, explained in Section 5.3) or a diffusers pipeline whose Hub count covers only the denoiser (†). Agreement allows for the 0-d scalars the Hub skips; NVFP4 and bitsandbytes releases are compared by logical count. *d* and layers are for the backbone. *Roles* is role accuracy against labels transferred from the corpus, where the model type is in the corpus.

| Repository | Storage | Params | Hub | | Family | *d* | Layers | Header (KB) | Roles |
|---|---|---|---|---|---|---|---|---|---|
| **Dense LM** (45) | | | | | | | | | |
| `Qwen/Qwen2.5-72B` | | 72.7 B | 72.7 B | = | dec | 8192 | 80 | 109 | 100 |
| `deepseek-ai/DeepSeek-R1-Distill-Llama-70B` | | 70.6 B | 70.6 B | = | dec | 8192 | 80 | 83 | |
| `unsloth/Llama-3.3-70B-Instruct` | | 70.6 B | 70.6 B | = | dec | 8192 | 80 | 83 | |
| `Qwen/QwQ-32B` | | 32.8 B | 32.8 B | = | dec | 5120 | 64 | 87 | 100 |
| `Qwen/Qwen3-32B` | | 32.8 B | 32.8 B | = | dec | 5120 | 64 | 80 | 100 |
| `unsloth/gemma-3-27b-it` | | 27.4 B | 27.4 B | = | prefix | 5376 | 62 | 163 | |
| `mistralai/Devstral-Small-2505` | | 23.6 B | 23.6 B | = | dec | 5120 | 40 | 41 | 100 |
| `mistralai/Mistral-Small-24B-Instruct-2501` | | 23.6 B | 23.6 B | = | dec | 5120 | 40 | 41 | 100 |
| `microsoft/phi-4` | | 14.7 B | 14.7 B | = | dec | 5120 | 40 | 28 | 100 |
| `mistralai/Mistral-Nemo-Instruct-2407` | | 12.2 B | 12.2 B | = | dec | 5120 | 40 | 41 | 100 |
| `THUDM/glm-4-9b-chat-hf` | | 9.40 B | 9.40 B | = | dec | 4096 | 40 | 50 | 100 |
| `unsloth/gemma-2-9b` | | 9.24 B | 9.24 B | = | dec | 3584 | 42 | 53 | 100 |
| `internlm/internlm3-8b-instruct` | | 8.80 B | 8.80 B | = | dec | 4096 | 48 | 49 | |
| `Qwen/Qwen3-8B` | | 8.19 B | 8.19 B | = | dec | 4096 | 36 | 45 | 100 |
| `ibm-granite/granite-3.3-8b-instruct` | | 8.17 B | 8.17 B | = | dec | 4096 | 40 | 41 | 100 |
| `swiss-ai/Apertus-8B-2509` | | 8.05 B | 8.05 B | = | dec | 4096 | 32 | 49 | 100 |
| `nvidia/Llama-3.1-Nemotron-Nano-8B-v1` | | 8.03 B | 8.03 B | = | dec | 4096 | 32 | 33 | |
| `unsloth/Meta-Llama-3.1-8B` | | 8.03 B | 8.03 B | = | dec | 4096 | 32 | 33 | |
| `Qwen/Qwen2.5-7B` | | 7.62 B | 7.62 B | = | dec | 3584 | 28 | 38 | 100 |
| `Qwen/Qwen2.5-Coder-7B-Instruct` | | 7.62 B | 7.62 B | = | dec | 3584 | 28 | 38 | 100 |
| `deepseek-ai/DeepSeek-R1-Distill-Qwen-7B` | | 7.62 B | 7.62 B | = | dec | 3584 | 28 | 38 | 100 |
| `tiiuae/Falcon3-7B-Base` | | 7.46 B | 7.46 B | = | dec | 3072 | 28 | 29 | |
| `allenai/OLMo-2-1124-7B` | | 7.30 B | 7.30 B | = | dec | 4096 | 32 | 40 | 100 |
| `unsloth/mistral-7b-v0.3` | | 7.25 B | 7.25 B | = | dec | 4096 | 32 | 33 | 98 |
| `mistralai/Mistral-7B-v0.1` | | 7.24 B | 7.24 B | = | dec | 4096 | 32 | 33 | 100 |
| `tiiuae/falcon-7b` | | 7.22 B | 7.22 B | = | dec | 4544 | 32 | 23 | 100 |
| `Qwen/Qwen3-4B` | | 4.02 B | 4.02 B | = | dec | 2560 | 36 | 45 | 100 |
| `microsoft/Phi-4-mini-instruct` | | 3.84 B | 3.84 B | = | dec | 3072 | 32 | 22 | 100 |
| `microsoft/Phi-3-mini-4k-instruct` | | 3.82 B | 3.82 B | = | dec | 3072 | 32 | 22 | 100 |
| `unsloth/Llama-3.2-3B` | | 3.21 B | 3.21 B | = | dec | 3072 | 28 | 29 | |
| `HuggingFaceTB/SmolLM3-3B` | | 3.08 B | 3.08 B | = | dec | 2048 | 36 | 37 | 100 |
| `microsoft/phi-2` | | 2.78 B | 2.78 B | = | dec | 2560 | 32 | 49 | 100 |
| `unsloth/gemma-2-2b` | | 2.61 B | 2.61 B | = | dec | 2304 | 26 | 33 | 100 |
| `HuggingFaceTB/SmolLM2-1.7B` | | 1.71 B | 1.71 B | = | dec | 2048 | 24 | 25 | |
| `openai-community/gpt2-xl` | | 1.61 B | 1.61 B | = | dec | 1600 | 48 | 57 | |
| `EleutherAI/pythia-1.4b` | | 1.52 B | 1.52 B | = | dec | 2048 | 24 | 41 | 92 |
| `facebook/opt-1.3b` | | 1.42 B | – | | dec | 2048 | 24 | 136 | 100 |
| `unsloth/Llama-3.2-1B` | | 1.24 B | 1.24 B | = | dec | 2048 | 16 | 16 | |
| `LiquidAI/LFM2-1.2B` | | 1.17 B | 1.17 B | = | enc | 2048 | 16 | 16 | 100 |
| `unsloth/gemma-3-1b-it` | | 1.00 B | 1.00 B | = | dec | 1152 | 26 | 38 | 100 |
| `Qwen/Qwen3-0.6B` | | 752 M | 752 M | = | dec | 1024 | 28 | 35 | 100 |
| `bigscience/bloom-560m` | | 559 M | 559 M | = | enc | 1024 | 24 | 30 | |
| `Qwen/Qwen2.5-0.5B` | | 494 M | 494 M | = | dec | 896 | 24 | 32 | 100 |
| `baidu/ERNIE-4.5-0.3B-PT` | | 361 M | 361 M | = | dec | 1024 | 18 | 18 | 100 |
| `openai-community/gpt2` | | 137 M | 137 M | = | dec | 768 | 12 | 14 | |
| **MoE LM** (24) | | | | | | | | | |
| `moonshotai/Kimi-K2-Instruct` | FP8 | 1.03 T | 1.03 T | = | dec | 7168 | 61 | 17,309 | |
| `deepseek-ai/DeepSeek-R1` | FP8 | 684 B | 684 B | = | dec | 7168 | 61 | 11,610 | |
| `deepseek-ai/DeepSeek-V3` | FP8 | 684 B | 684 B | = | dec | 7168 | 61 | 11,610 | |
| `deepseek-ai/DeepSeek-V3.1` | FP8 | 684 B | 685 B | ≠ | dec | 7168 | 61 | 11,610 | |
| `meituan-longcat/LongCat-Flash-Chat` | | 562 B | 562 B | = | dec | 6144 | 28 | 5,398 | |
| `Qwen/Qwen3-Coder-480B-A35B-Instruct` | | 480 B | 480 B | = | dec | 6144 | 62 | 3,702 | |
| `zai-org/GLM-4.5` | | 358 B | 358 B | = | dec | 5120 | 92 | 5,454 | |
| `Qwen/Qwen3-235B-A22B` | | 235 B | 235 B | = | dec | 4096 | 94 | 4,498 | |
| `MiniMaxAI/MiniMax-M2` | FP8 | 229 B | 229 B | = | dec | 3072 | 62 | 12,338 | |
| `openai/gpt-oss-120b` | MXFP4 | 117 B | 117 B | = | dec | 2880 | 36 | 76 | 100 |
| `zai-org/GLM-4.5-Air` | | 110 B | 110 B | = | dec | 4096 | 46 | 2,229 | |
| `unsloth/Llama-4-Scout-17B-16E-Instruct` | | 109 B | 109 B | = | prefix | 5120 | 48 | 148 | |

| Repository | Storage | Params | Hub | | Family | $d$ | Layers | Header (KB) | Roles |
|---|---|---|---|---|---|---|---|---|---|
| Qwen/Qwen3-Next-80B-A3B-Instruct | | 81.3 B | 81.3 B | = | dec | 2048 | 48 | 9,215 | |
| tencent/Hunyuan-A13B-Instruct | | 80.4 B | 80.4 B | = | dec | 4096 | 32 | 792 | |
| mistralai/Mixtral-8x7B-v0.1 | | 46.7 B | 46.7 B | = | dec | 4096 | 32 | 125 | |
| microsoft/Phi-3.5-MoE-instruct | | 41.9 B | 41.9 B | = | dec | 4096 | 32 | 244 | |
| Qwen/Qwen3-30B-A3B | | 30.5 B | 30.5 B | = | dec | 2048 | 48 | 2,276 | |
| baidu/ERNIE-4.5-21B-A3B-PT | | 21.9 B | 21.9 B | = | dec | 2560 | 28 | 667 | |
| openai/gpt-oss-20b | MXFP4 | 20.9 B | 20.9 B | = | dec | 2880 | 24 | 51 | 100 |
| moonshotai/Moonlight-16B-A3B | | 16.0 B | 16.0 B | = | dec | 2048 | 27 | 640 | |
| deepseek-ai/DeepSeek-V2-Lite | | 15.7 B | 15.7 B | = | dec | 2048 | 27 | 643 | |
| Qwen/Qwen1.5-MoE-A2.7B | | 14.3 B | 14.3 B | = | dec | 2048 | 24 | 563 | |
| allenai/OLMoE-1B-7B-0924 | | 6.92 B | 6.92 B | = | dec | 2048 | 16 | 389 | |
| ibm-granite/granite-3.1-3b-a800m-instruct | | 3.30 B | 3.30 B | = | dec | 1536 | 32 | 34 | 100 |
| **SSM / hybrid** (9) | | | | | | | | | |
| nvidia/NVIDIA-Nemotron-Nano-9B-v2 | | 8.89 B | 8.89 B | = | dec | 4480 | 56 | 37 | |
| nvidia/Nemotron-H-8B-Base-8K | | 8.10 B | 8.10 B | = | dec | 4096 | 52 | 34 | |
| tiiuae/Falcon-H1-7B-Base | | 7.59 B | 7.59 B | = | dec | 3072 | 44 | 83 | 100 |
| tiiuae/falcon-mamba-7b | | 7.27 B | 7.27 B | = | dec | 4096 | 64 | 70 | 100 |
| ibm-granite/granite-4.0-h-tiny | | 6.94 B | 6.94 B | = | dec | 1536 | 40 | 66 | |
| NX-AI/xLSTM-7b | | 6.87 B | 6.87 B | = | dec | 4096 | 32 | 56 | 37 |
| state-spaces/mamba-2.8b-hf | | 2.77 B | 2.77 B | = | dec | 2560 | 64 | 69 | |
| Zyphra/Zamba2-2.7B | | 2.66 B | 2.66 B | = | enc | 128 | 5 | 58 | 4 |
| RWKV/v6-Finch-1B6-HF | | 1.60 B | – | | dec | 2048 | 24 | 142 | |
| **Encoder** (7) | | | | | | | | | |
| FacebookAI/xlm-roberta-large | | 561 M | 561 M | = | enc | 1024 | 24 | 48 | |
| answerdotai/ModernBERT-large | | 396 M | 396 M | = | enc | 1024 | 28 | 18 | 100 |
| FacebookAI/roberta-large | | 355 M | 355 M | = | enc | 1024 | 24 | 47 | 100 |
| microsoft/deberta-v3-base | | 186 M | – | | enc | 768 | 12 | 106 | 100 |
| google-bert/bert-base-uncased | | 110 M | 110 M | = | enc | 768 | 12 | 23 | 100 |
| distilbert/distilbert-base-uncased | | 67.0 M | 67.0 M | = | enc | 768 | 6 | 12 | 100 |
| albert/albert-base-v2 | | 11.8 M | 11.8 M | = | ? | – | – | 4 | 45 |
| **Embedding** (10) | | | | | | | | | |
| Qwen/Qwen3-Embedding-8B | | 7.57 B | 7.57 B | = | dec | 4096 | 36 | 43 | |
| Qwen/Qwen3-Embedding-0.6B | | 596 M | 596 M | = | dec | 1024 | 28 | 33 | |
| Qwen/Qwen3-Reranker-0.6B | | 596 M | 596 M | = | dec | 1024 | 28 | 34 | 100 |
| BAAI/bge-m3 | | 568 M | – | | enc | 1024 | 24 | 138 | |
| intfloat/multilingual-e5-large | | 560 M | 560 M | = | enc | 1024 | 24 | 44 | |
| BAAI/bge-large-en-v1.5 | | 335 M | 335 M | = | enc | 1024 | 24 | 44 | |
| Alibaba-NLP/gte-multilingual-base | | 305 M | 305 M | = | enc | 768 | 12 | 15 | |
| nomic-ai/nomic-embed-text-v1.5 | | 137 M | 137 M | = | enc | 768 | 12 | 11 | |
| sentence-transformers/all-mpnet-base-v2 | | 109 M | 109 M | = | enc | 768 | 12 | 21 | |
| sentence-transformers/all-MiniLM-L6-v2 | | 22.7 M | 22.7 M | = | enc | 384 | 6 | 11 | |
| **Seq2seq** (6) | | | | | | | | | |
| facebook/nllb-200-distilled-600M | | 1.40 B | – | | enc-dec | 1024 | 12 | 160 | 100 |
| google/mt5-base | | 967 M | – | | enc-dec | 768 | 12 | 155 | 100 |
| facebook/m2m100_418M | | 877 M | – | | enc-dec | 1024 | 12 | 171 | 100 |
| google/flan-t5-large | | 783 M | 783 M | = | enc-dec | 1024 | 24 | 67 | 100 |
| facebook/bart-large-cnn | | 406 M | 406 M | = | enc-dec | 1024 | 12 | 59 | 100 |
| google-t5/t5-base | | 223 M | 223 M | = | enc-dec | 768 | 12 | 30 | 100 |
| **Speech / audio** (23) | | | | | | | | | |
| Qwen/Qwen2.5-Omni-7B | | 10.7 B | 10.7 B | = | diff | 3584 | 28 | 301 | |
| Qwen/Qwen2-Audio-7B-Instruct | | 8.40 B | 8.40 B | = | prefix | 4096 | 32 | 104 | |
| kyutai/stt-2.6b-en | | 2.71 B | 2.62 B | $\neq$ | prefix | 2048 | 48 | 75 | |
| microsoft/VibeVoice-1.5B | | 2.70 B | 2.70 B | = | diff | 1536 | 28 | 155 | |
| ResembleAI/chatterbox | | 2.67 B | – | | prefix | 1024 | 30 | 749 | |
| facebook/seamless-m4t-v2-large | | 2.31 B | 2.31 B | = | enc-dec | 1024 | 24 | 272 | 96 |
| nari-labs/Dia-1.6B-0626 | | 1.61 B | 1.61 B | = | enc-dec | 2048 | 18 | 41 | 100 |
| openai/whisper-large-v3 | | 1.54 B | 1.54 B | = | enc-dec | 1280 | 32 | 293 | |
| openai/whisper-large-v3-turbo | | 809 M | 809 M | = | enc-dec | 1280 | 4 | 67 | 100 |
| distil-whisper/distil-large-v3 | | 756 M | 756 M | = | enc-dec | 1280 | 2 | 123 | |
| facebook/musicgen-small | | 591 M | 591 M | = | enc-dec | 1024 | 24 | 77 | |
| facebook/wav2vec2-large-xlsr-53 | | 317 M | – | | enc | 1024 | 24 | 181 | 100 |
| facebook/hubert-large-ls960-ft | | 315 M | – | | enc | 1024 | 24 | 178 | 100 |
| microsoft/wavlm-large | | 315 M | – | | enc | 1024 | 24 | 178 | |
| openai/whisper-small | | 242 M | 242 M | = | enc-dec | 768 | 12 | 54 | 100 |
| laion/clap-htsat-unfused | | 154 M | – | | enc | 768 | 12 | 157 | 100 |
| microsoft/speecht5_tts | | 146 M | – | | enc-dec | 768 | 12 | 146 | 100 |

| Repository | Storage | Params | Hub | | Family | $d$ | Layers | Header (KB) | Roles |
|---|---|---|---|---|---|---|---|---|---|
| facebook/wav2vec2-base-960h | | 94.4 M | 94.4 M | = | enc | 768 | 12 | 25 | 100 |
| MIT/ast-finetuned-audioset-10-10-0.4593 | | 86.6 M | 86.6 M | = | enc | 768 | 12 | 28 | |
| UsefulSensors/moonshine-base | | 61.5 M | 61.5 M | = | enc-dec | 416 | 8 | 24 | 100 |
| openai/whisper-tiny | | 37.8 M | 37.8 M | = | enc-dec | 384 | 4 | 19 | 100 |
| facebook/encodec_24khz | | 23.3 M | 23.3 M | = | prefix | 64 | 4 | 26 | |
| speechbrain/spkrec-ecapa-voxceleb | | 20.8 M | – | | enc | 128 | 7 | 126 | |
| **Vision** (19) | | | | | | | | | |
| google/siglip-so400m-patch14-384 | | 878 M | 878 M | = | prefix | 1152 | 27 | 106 | 96 |
| facebook/sam-vit-huge | | 641 M | 641 M | = | enc-dec | 1280 | 32 | 67 | 100 |
| openai/clip-vit-large-patch14 | | 428 M | 428 M | = | prefix | 768 | 12 | 70 | |
| google/siglip2-base-patch16-224 | | 375 M | 375 M | = | prefix | 768 | 12 | 48 | 92 |
| Intel/dpt-large | | 342 M | 342 M | = | enc | 1024 | 24 | 54 | |
| depth-anything/Depth-Anything-V2-Large-hf | | 335 M | 335 M | = | enc | 1024 | 24 | 59 | |
| facebook/dinov2-large | | 304 M | 304 M | = | enc | 1024 | 24 | 47 | |
| IDEA-Research/grounding-dino-base | | 233 M | 233 M | = | enc-dec | 512 | 24 | 163 | |
| facebook/sam2.1-hiera-large | | 224 M | 224 M | = | enc-dec | 144 | 48 | 106 | 95 |
| facebook/mask2former-swin-large-coco-panoptic | | 216 M | – | | enc-dec | 768 | 24 | 240 | |
| google/owlv2-base-patch16-ensemble | | 155 M | 155 M | = | prefix | 512 | 12 | 50 | |
| facebook/convnext-base-224 | | 88.6 M | – | | enc | 512 | 36 | 145 | |
| microsoft/swin-base-patch4-window7-224 | | 87.8 M | 87.8 M | = | enc | 512 | 24 | 55 | |
| google/vit-base-patch16-224 | | 86.6 M | 86.6 M | = | enc | 768 | 12 | 23 | |
| nvidia/segformer-b5-finetuned-ade-640-640 | | 84.7 M | – | | enc | 320 | 53 | 366 | |
| PekingU/rtdetr_r50vd | | 43.0 M | 43.0 M | = | enc-dec | 256 | 48 | 96 | 84 |
| facebook/detr-resnet-50 | | 41.6 M | 41.6 M | = | enc-dec | 256 | 13 | 62 | |
| microsoft/resnet-50 | | 25.6 M | 25.6 M | = | enc | 256 | 48 | 41 | 100 |
| timm/efficientnet_b0.ra_in1k | | 5.33 M | 5.33 M | = | enc | 480 | 16 | 32 | |
| **Vision–language** (18) | | | | | | | | | |
| Qwen/Qwen3-VL-235B-A22B-Instruct | | 236 B | 236 B | = | prefix | 4096 | 94 | 174 | |
| zai-org/GLM-4.5V | | 108 B | 108 B | = | prefix | 4096 | 46 | 2,464 | |
| Qwen/Qwen2.5-VL-72B-Instruct | | 73.4 B | 73.4 B | = | prefix | 8192 | 80 | 151 | |
| Qwen/Qwen3-VL-30B-A3B-Instruct | | 31.1 B | 31.1 B | = | prefix | 2048 | 48 | 107 | |
| moonshotai/Kimi-VL-A3B-Instruct | | 16.4 B | 16.4 B | = | prefix | 2048 | 27 | 765 | |
| nvidia/NVIDIA-Nemotron-Nano-12B-v2-VL-BF16 | | 13.2 B | 13.2 B | = | prefix | 5120 | 62 | 95 | |
| mistral-community/pixtral-12b | | 12.7 B | 12.7 B | = | prefix | 5120 | 40 | 76 | |
| unsloth/Llama-3.2-11B-Vision-Instruct | | 10.7 B | 10.7 B | = | enc-dec | 4096 | 40 | 116 | |
| Qwen/Qwen2.5-VL-7B-Instruct | | 8.29 B | 8.29 B | = | prefix | 3584 | 28 | 79 | |
| llava-hf/llava-onevision-qwen2-7b-ov-hf | | 8.03 B | 8.03 B | = | prefix | 3584 | 28 | 99 | |
| allenai/Molmo-7B-D-0924 | | 8.02 B | 8.02 B | = | prefix | 3584 | 28 | 79 | |
| OpenGVLab/InternVL3-8B-hf | | 7.94 B | 7.94 B | = | prefix | 3584 | 28 | 97 | |
| llava-hf/llava-1.5-7b-hf | | 7.06 B | 7.06 B | = | prefix | 4096 | 32 | 90 | |
| microsoft/Phi-4-multimodal-instruct | | 6.40 B | 5.57 B | ≠ | enc-dec | 3072 | 32 | 368 | |
| deepseek-ai/Janus-Pro-7B | | 4.99 B | – | | prefix | 4096 | 20 | 249 | |
| unsloth/gemma-3-4b-it | | 4.30 B | 4.30 B | = | prefix | 2560 | 34 | 116 | |
| HuggingFaceTB/SmolVLM2-2.2B-Instruct | | 2.25 B | 2.25 B | = | prefix | 2048 | 24 | 83 | |
| lerobot/smolvla_base | | 450 M | 450 M | ≠ | prefix | 960 | 16 | 72 | |
| **Image / video gen.** (14) | | | | | | | | | |
| Qwen/Qwen-Image | | 28.8 B | 20.4 B | † | diff | 3584 | 28 | 321 | |
| HiDream-ai/HiDream-I1-Full | | 22.8 B | 17.1 B | † | prefix | 4096 | 24 | 338 | |
| hunyuanvideo-community/HunyuanVideo | | 20.7 B | 12.8 B | † | diff | 4096 | 32 | 229 | |
| Wan-AI/Wan2.2-TI2V-5B-Diffusers | | 11.4 B | 5.00 B | † | prefix | 4096 | 24 | 134 | |
| Wan-AI/Wan2.1-T2V-1.3B-Diffusers | | 7.23 B | 1.42 B | † | prefix | 4096 | 24 | 132 | |
| Lightricks/LTX-Video | | 7.10 B | 1.92 B | † | diff | 4096 | 24 | 2,979 | |
| THUDM/CogVideoX-2b | | 6.67 B | 1.69 B | † | prefix | 4096 | 24 | 159 | |
| PixArt-alpha/PixArt-Sigma-XL-2-1024-MS | | 5.46 B | 611 M | † | diff | 4096 | 24 | 120 | |
| Efficient-Large-Model/Sana_1600M_1024px_diffusers | | 5.32 B | – | | diff | 2304 | 26 | 282 | |
| playgroundai/playground-v2.5-1024px-aesthetic | | 3.47 B | 2.57 B | † | enc-dec | 1280 | 32 | 1,371 | |
| stabilityai/sdxl-turbo | | 3.47 B | 2.57 B | † | enc-dec | 1280 | 32 | 1,448 | |
| stabilityai/stable-diffusion-xl-base-1.0 | | 3.47 B | 2.57 B | † | enc-dec | 1280 | 32 | 1,853 | |
| stable-diffusion-v1-5/stable-diffusion-v1-5 | | 1.37 B | 860 M | † | enc-dec | 768 | 12 | 846 | |
| facebook/DiT-XL-2-256 | | 833 M | – | | enc | 1152 | 28 | 346 | |
| **Time series / tabular** (6) | | | | | | | | | |
| google/timesfm-2.0-500m-pytorch | | 499 M | 499 M | = | dec | 1280 | 50 | 89 | |
| amazon/chronos-bolt-base | | 205 M | 205 M | = | enc-dec | 768 | 12 | 31 | 100 |
| Salesforce/moirai-1.1-R-base | | 91.4 M | 91.4 M | = | enc | 768 | 12 | 18 | |
| amazon/chronos-t5-small | | 46.2 M | 46.2 M | = | enc-dec | 512 | 6 | 15 | 95 |
| Prior-Labs/TabPFN-v2-clf | | 7.24 M | – | | enc | 768 | 12 | 92 | |

| Repository | Storage | Params | Hub | | Family | $d$ | Layers | Header (KB) | Roles |
|---|---|---|---|---|---|---|---|---|---|
| `ibm-granite/granite-timeseries-ttm-r2` | | 805 K | 805 K | = | enc | 48 | 6 | 19 | |
| **Quantized release** (10) | | | | | | | | | |
| `nvidia/DeepSeek-R1-0528-FP4` | NVFP4 | 684 B | 397 B | ≠ | dec | 7168 | 61 | 20,848 | |
| `Qwen/Qwen3-30B-A3B-GPTQ-Int4` | GPTQ | 30.5 B | 30.5 B | = | dec | 2048 | 48 | 8,867 | |
| `Qwen/Qwen3-8B-FP8` | FP8 | 8.19 B | 8.19 B | = | dec | 4096 | 36 | 76 | 100 |
| `mlx-community/Qwen3-8B-4bit` | MLX | 8.19 B | 8.19 B | = | dec | 4096 | 36 | 101 | |
| `nvidia/Qwen3-8B-FP4` | NVFP4 | 8.19 B | 4.72 B | ≠ | dec | 4096 | 36 | 131 | |
| `unsloth/Qwen3-8B-bnb-4bit` | NF4 | 8.19 B | 8.41 B | ≠ | dec | 4096 | 36 | 193 | |
| `RedHatAI/Meta-Llama-3.1-8B-Instruct-FP8` | FP8 | 8.03 B | 8.03 B | = | dec | 4096 | 32 | 83 | |
| `nvidia/Llama-3.1-8B-Instruct-FP4` | NVFP4 | 8.03 B | 4.54 B | ≠ | dec | 4096 | 32 | 109 | |
| `Qwen/Qwen2.5-7B-Instruct-AWQ` | AWQ | 7.62 B | 7.62 B | = | dec | 3584 | 28 | 80 | |
| `unsloth/Llama-3.2-1B-Instruct-bnb-4bit` | NF4 | 1.24 B | 1.27 B | ≠ | dec | 2048 | 16 | 81 | |

# G Additional interface views

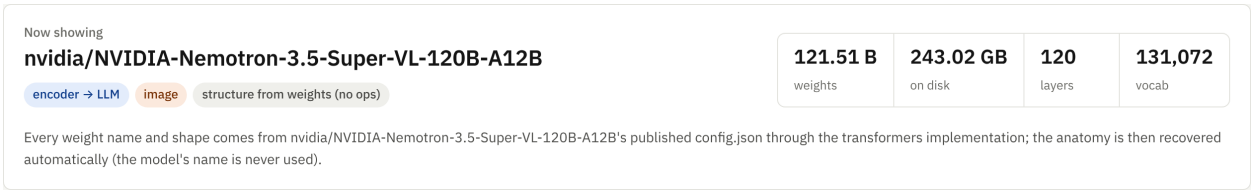



**Figure 12:** Summary strip for Nemotron 3.5 Super VL: weights, file size, layers and vocabulary, with the family and modality tags.

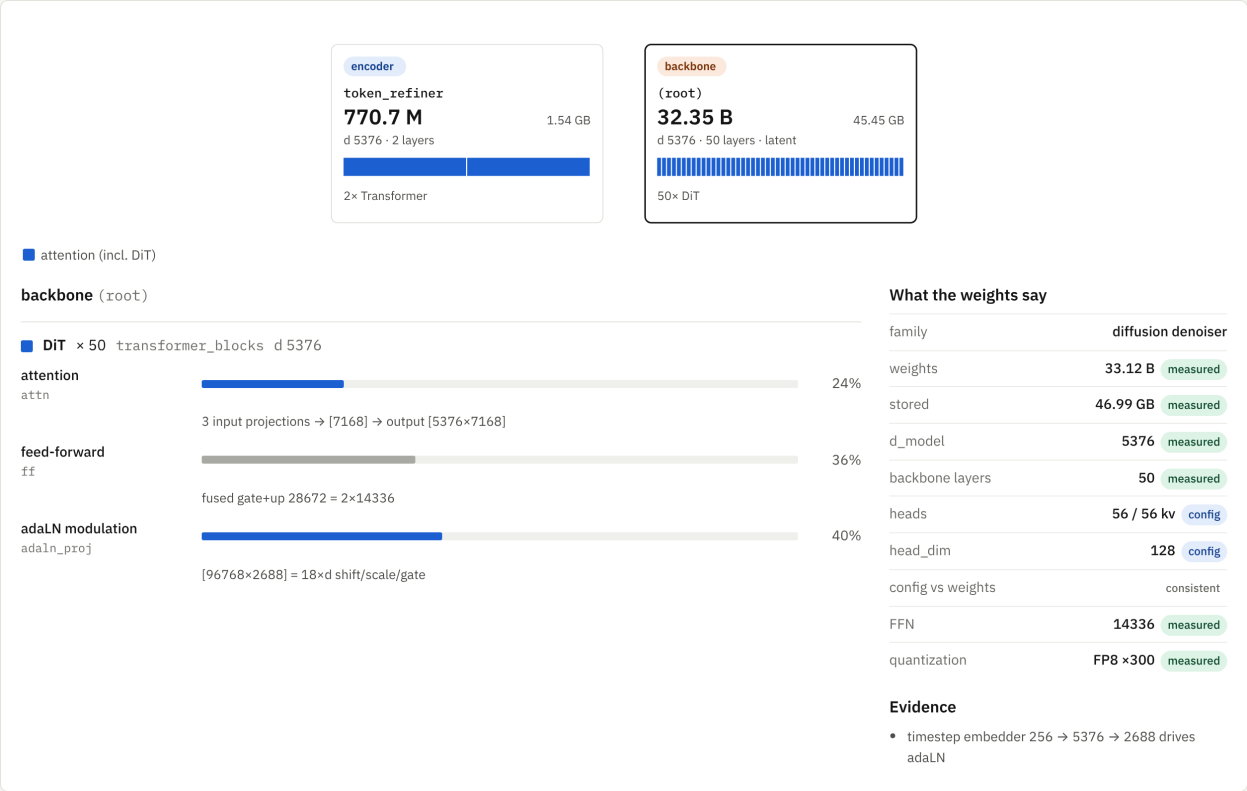



**Figure 13:** Anatomy view of the FastH3 Omni FP8 denoiser (reconstructed manifest): a two-block token refiner and a 50-block DiT backbone. The head geometry is measured from per-head norms, and the FP8 storage of 300 tensors is shown with logical and stored sizes.

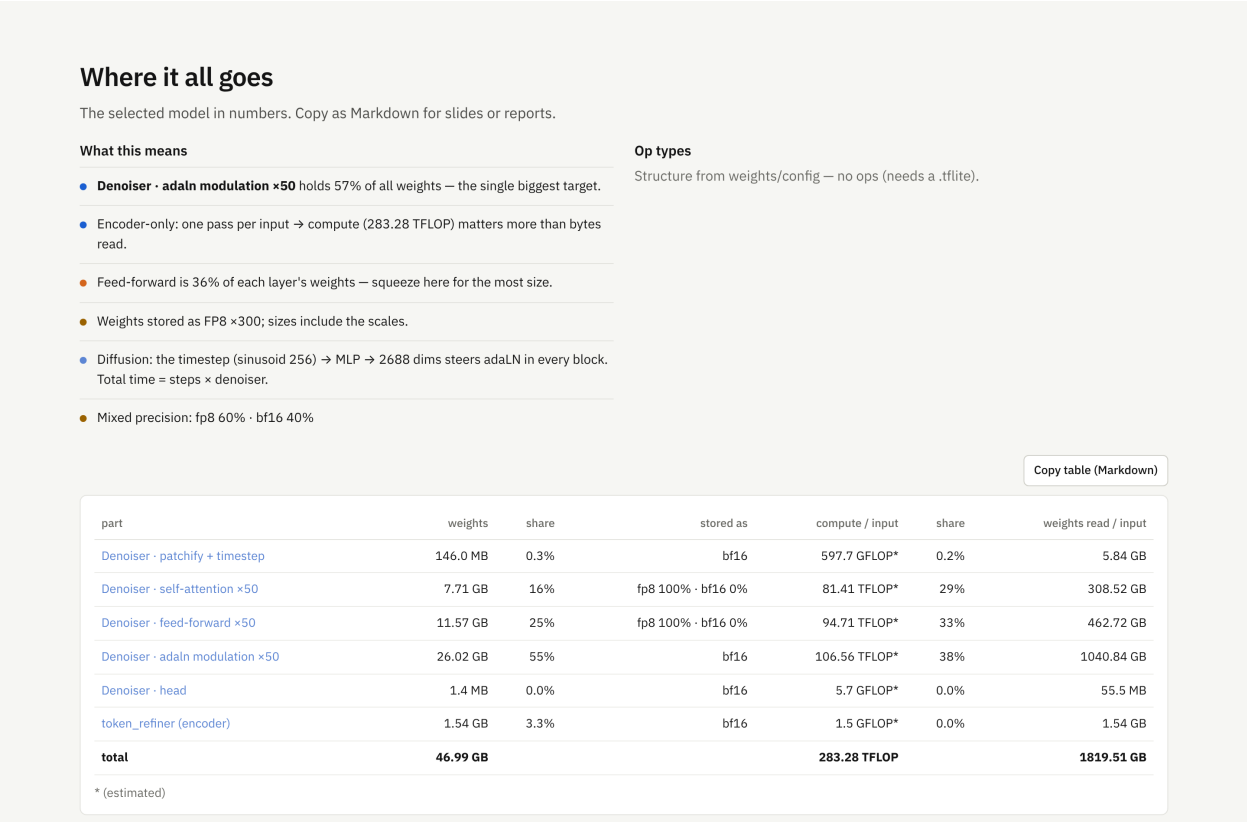



**Figure 14:** Report view for the same model: where parameters, compute and memory traffic go, with generated observations.